Computer Organisation & Program Execution 2019

5

Asynchronism

Uwe R. Zimmer - The Australian National University

References for this chapter

[Patterson17]

David A. Patterson & John L. Hennessy
Computer Organization and Design – The Hardware/Software Interface
Chapter 4 "The Processor",
Chapter 6 "Parallel Processors from Client to Cloud"
ARM edition, Morgan Kaufmann 2017

Why?

How do you handle your communication flow?

Why?

How do you handle your communication flow?

☞ Do you have times when you check certain communication?

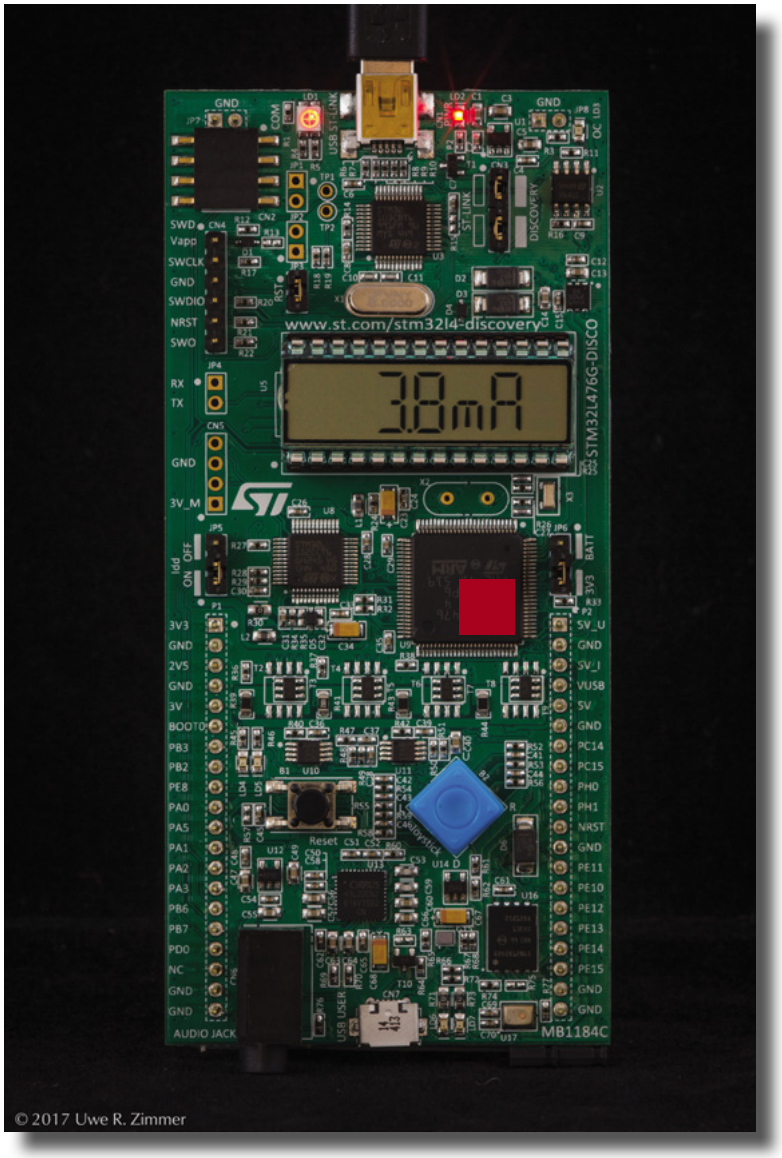
☞ Is certain communication interrupting you? – at any time?

☞ Do you assign "importance levels" to your communication channels/sources?

STM32L476 Discovery

CPU

… running its sequence of machine instructions.



STM32L476 Discovery

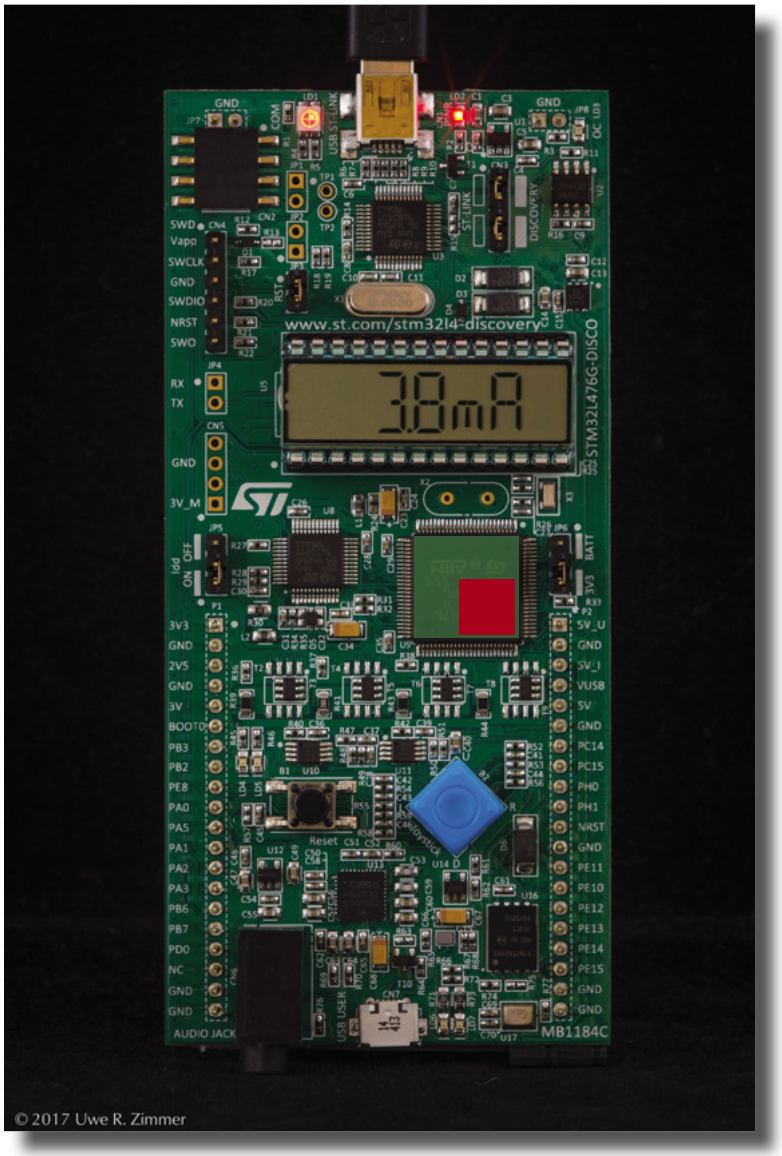
CPU

… running its sequence of machine instructions.

☞ How to interact with all the other devices inside the

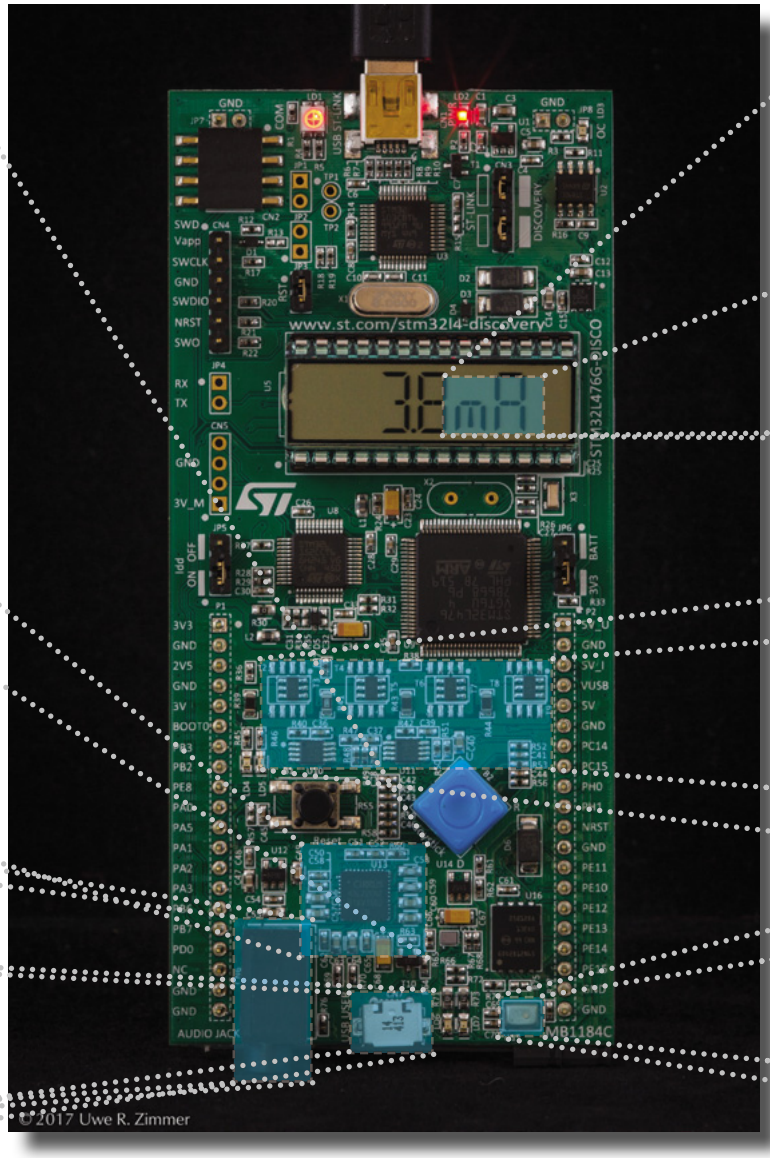
MCU

?



STM32L476 Discovery

Multiplexed 24 bit
ΣΔ-DAConverter with
stereo power amp



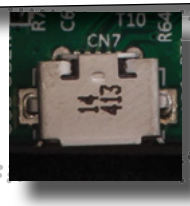
"9 axis" motion sensor
(underneath display):

- 3 axis accelerometer
- 3 axis gyroscope
- 3 axis magnetometer

Current meter to MCU
60 nA … 50 mA

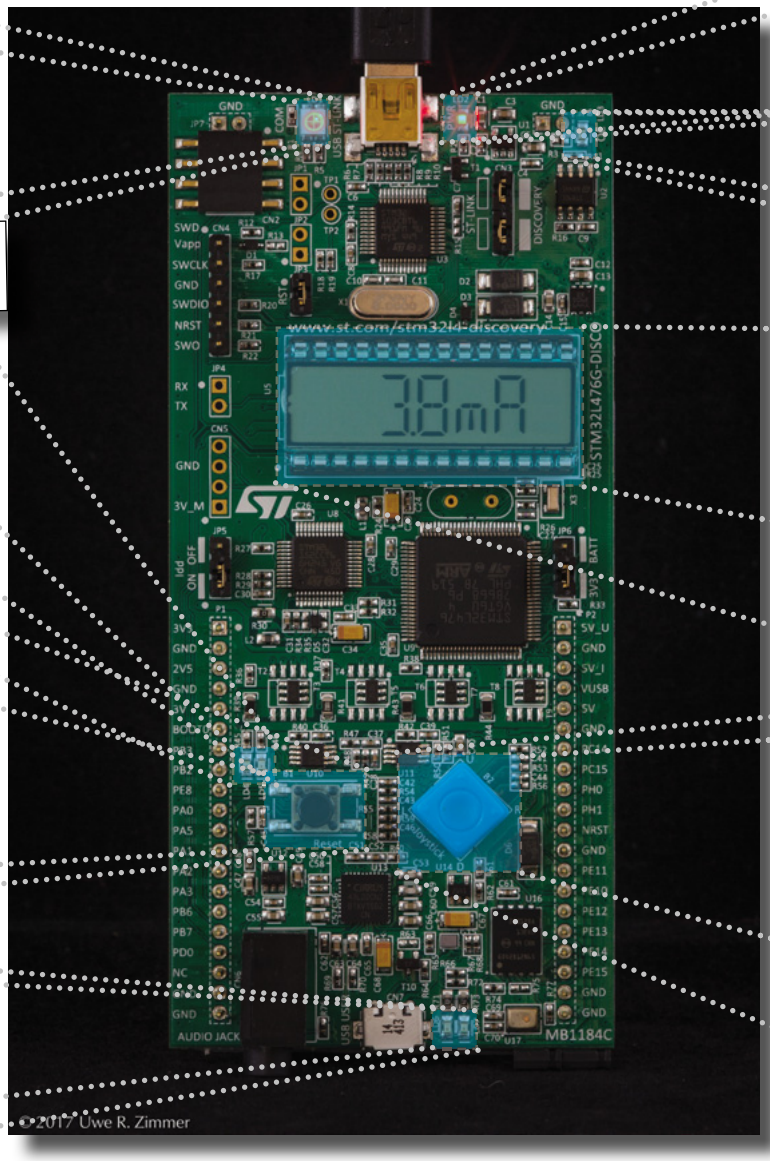
Headphone jack

USB OTG

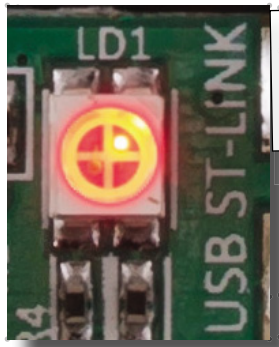


Microphone

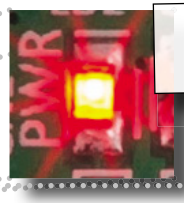
STM32L476 Discovery



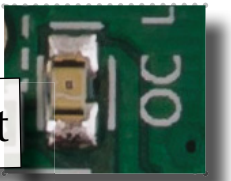
Debugger state



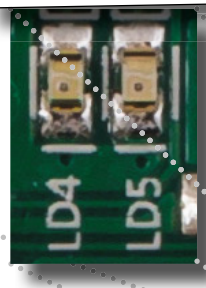
Power



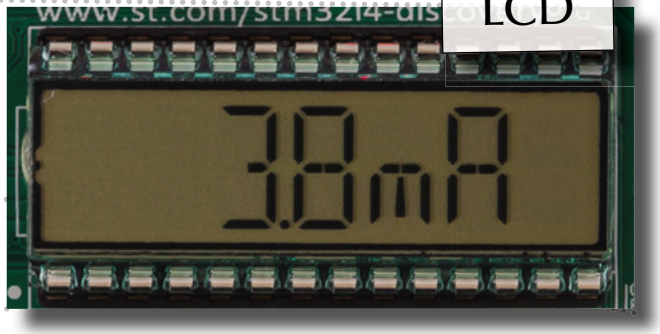
Over current



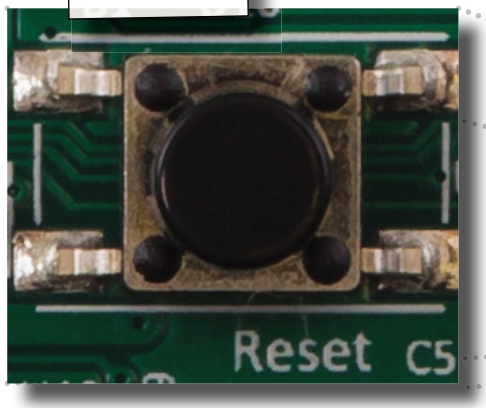
User LEDs



LCD

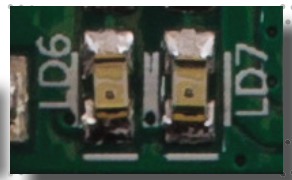


Reset



Joystick

OTG LEDs



STM32L476 Discovery

CPU

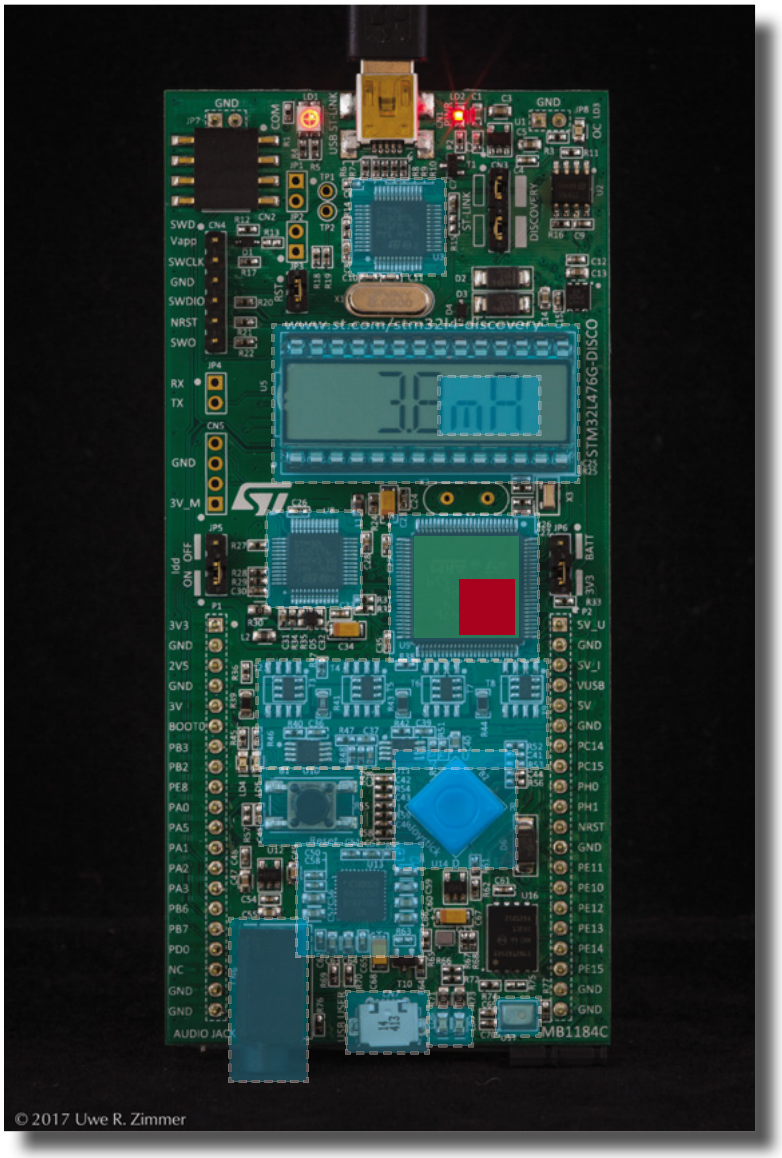
… running its sequence of machine instructions.

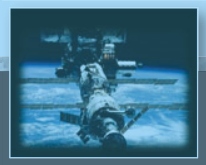
☞ How to interact with all the other devices inside the

MCU

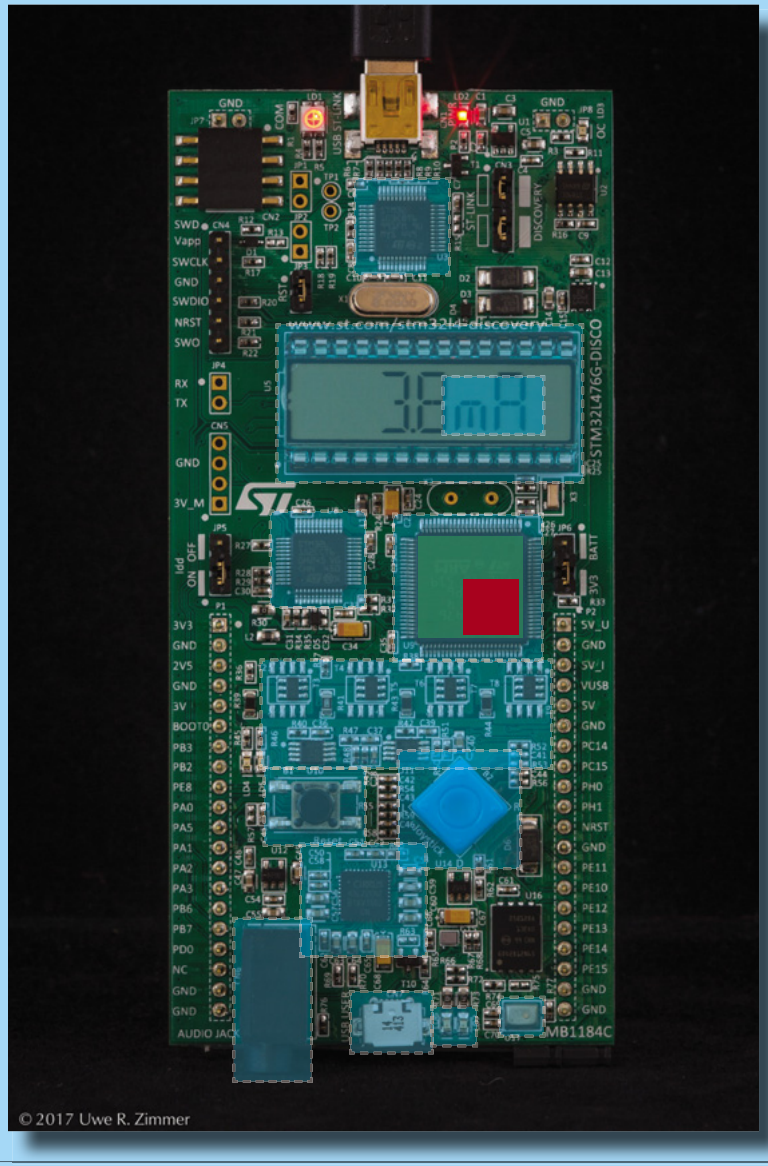
?

☞ … and then with all the devices on the board?





STM32L476 Discovery



CPU

… running its sequence of machine instructions.

☞ How to interact with all the other devices inside the

MCU

?

☞ … and then with all the devices on the board?

☞ and then with the

rest of the world

… which is connected to the board?

Polling

Sequential machine instructions

- ☞ All external devices need to be “checked” by asking for their status.
- ☞ This should usually happen (semi-) regularly.

Polling

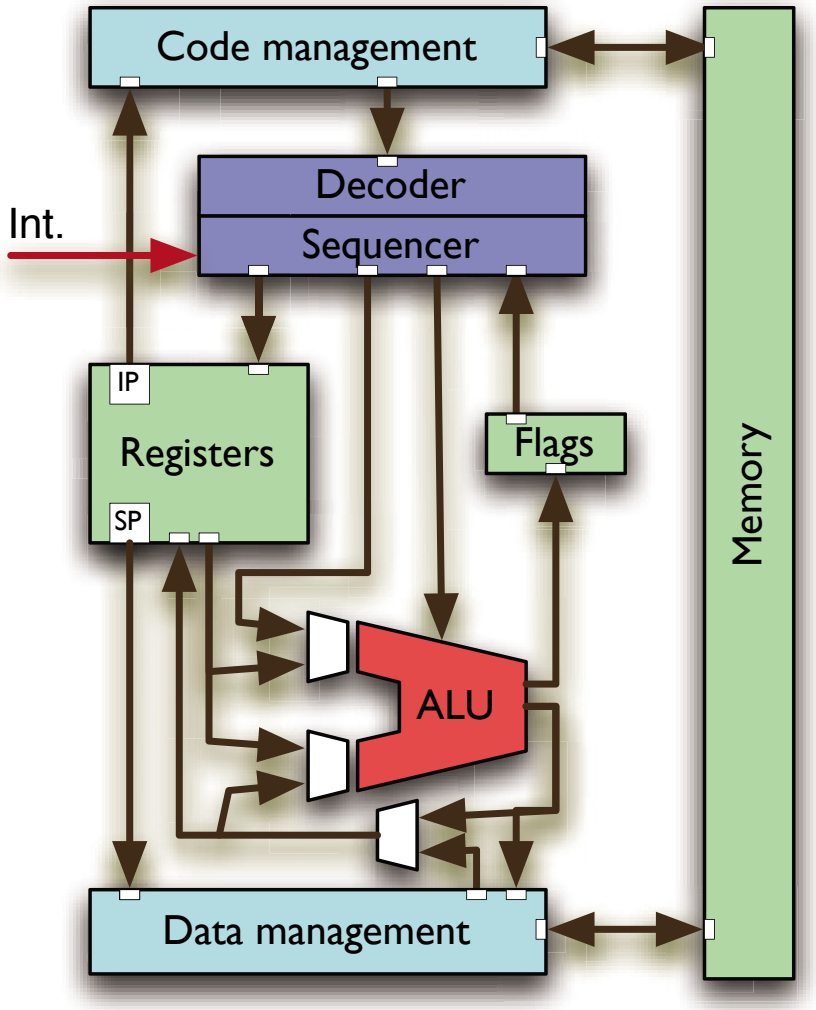
Sequential machine instructions

- ☞ All external devices need to be “checked” by asking for their status.
- ☞ This should usually happen (semi-) regularly.

☞ This will lead to a loop of **polling** requests.

- ↗ Maximal latencies can be calculated straight forward.
- ↗ Simplicity of design (with small number of devices).
- ↗ Fastest option with small number of devices (like: one).
- ↘ All devices will need to wait their turn
  … even if this device is the only one with new data!
- ↘ The “main” program transforms into one large loop which can be hard to handle in terms of scalable program design.
- ↘ Events or data can be missed.

Processor Architectures



Interrupts

- One or multiple lines wired directly into the sequencer

☞ *Required for*:
Pre-emptive scheduling, **Timer driven actions**, **Transient hardware interactions**, …

☞ Usually preceded by an external logic ("interrupt controller") which accumulates and encodes all external requests.

On interrupt (if unmasked):

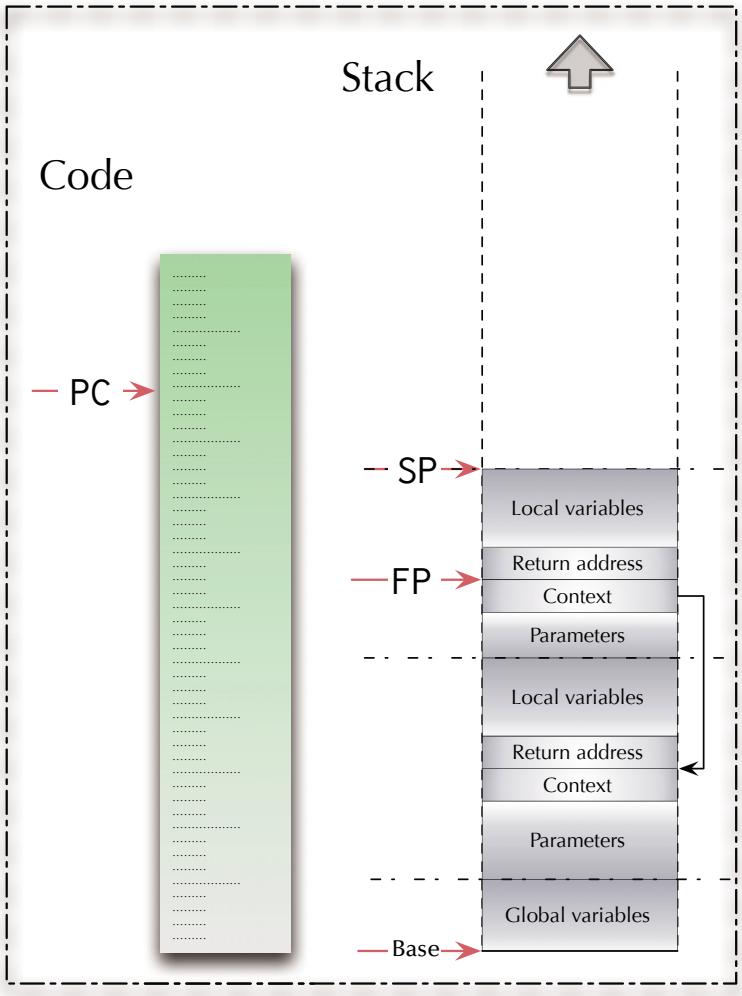
- CPU stops normal sequencer flow.
- Lookup of interrupt handler's address
- Current IP and state pushed onto stack.
- IP set to interrupt handler.

Bahia Honda Rail Bridge (Creative Commons Attribution-ShareAlike 3.0, Photography by MrX at English Wikipedia)

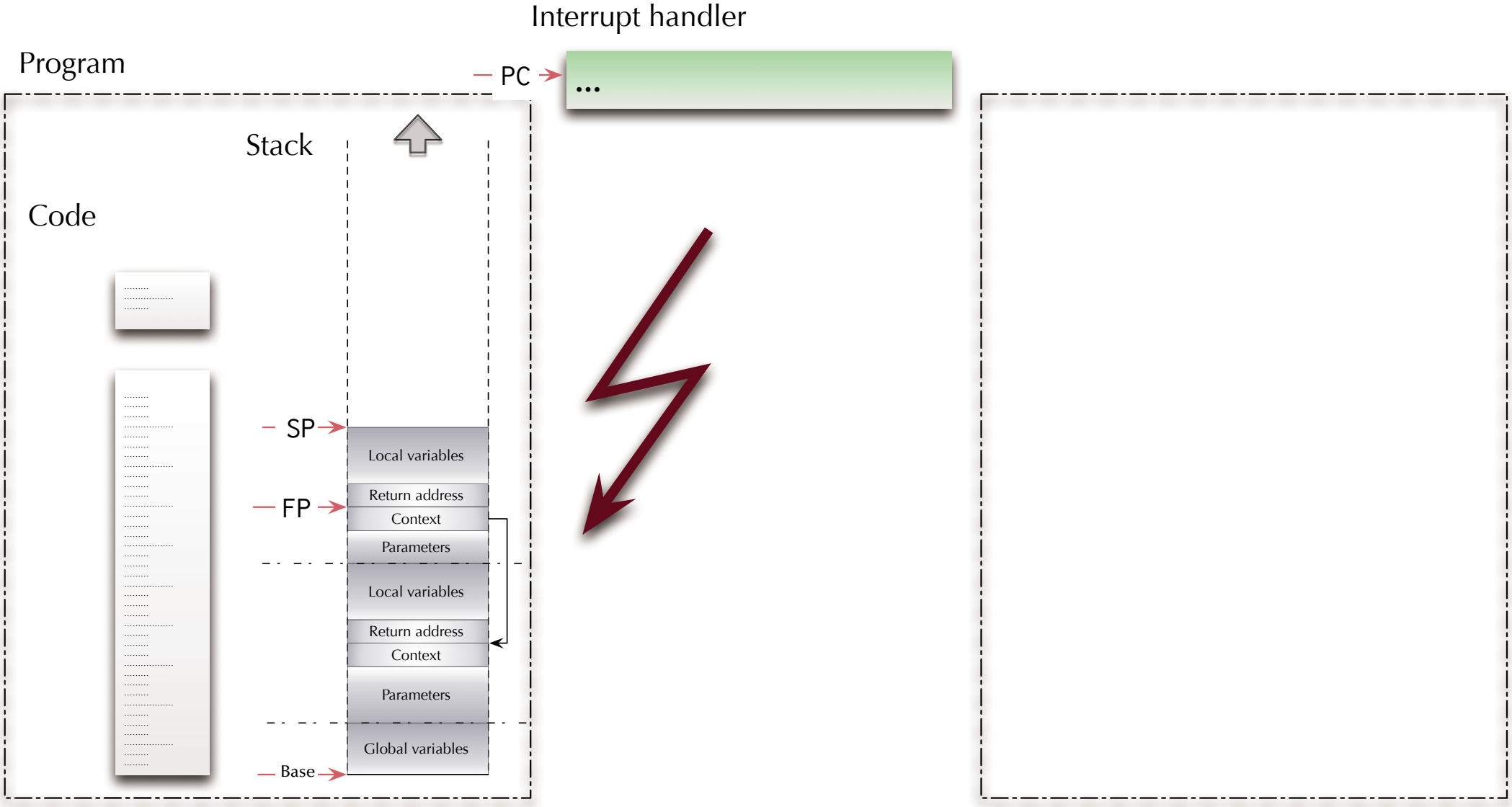
Interrupt processing

Interrupt handler

Program



Interrupt processing



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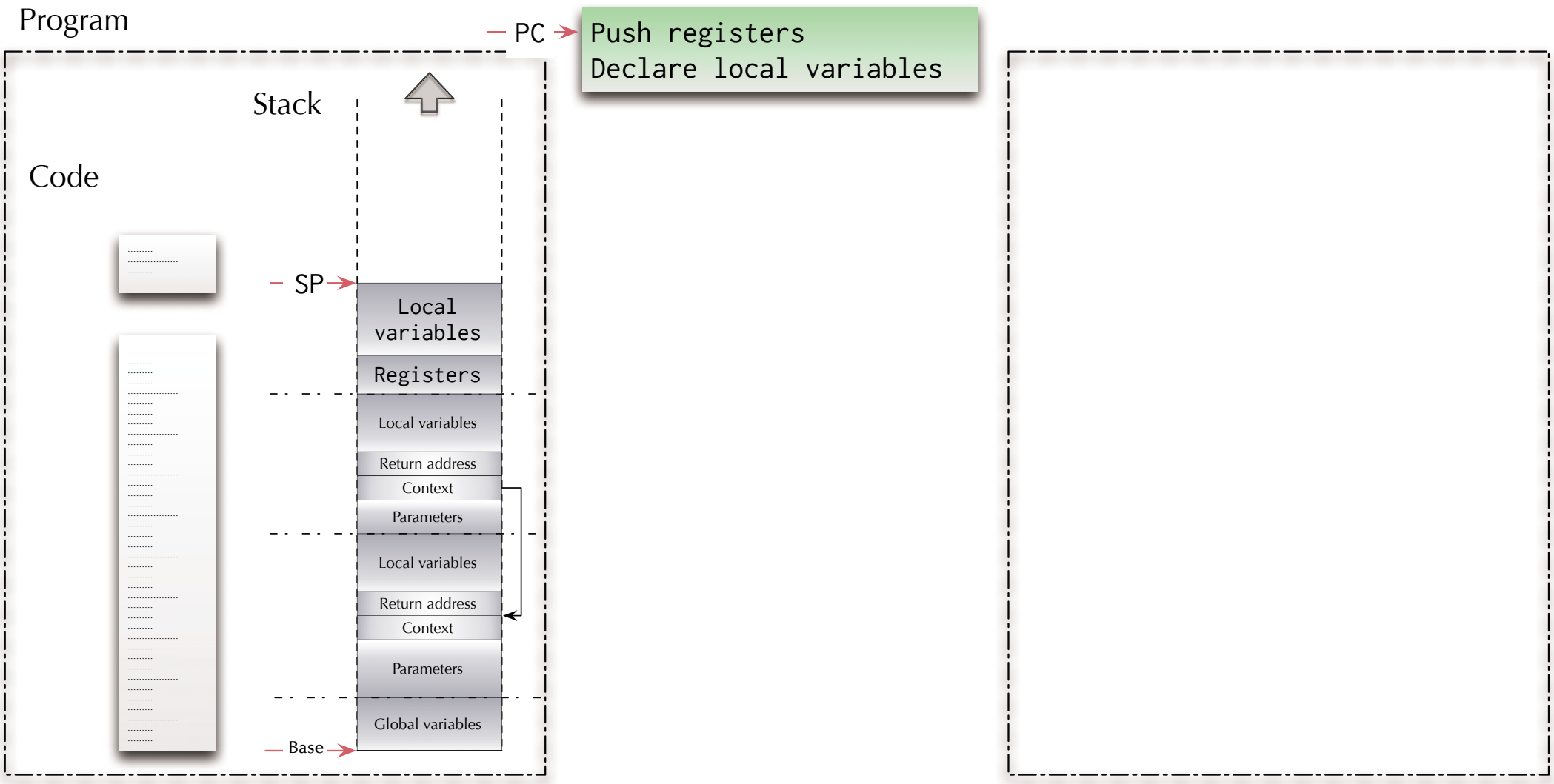

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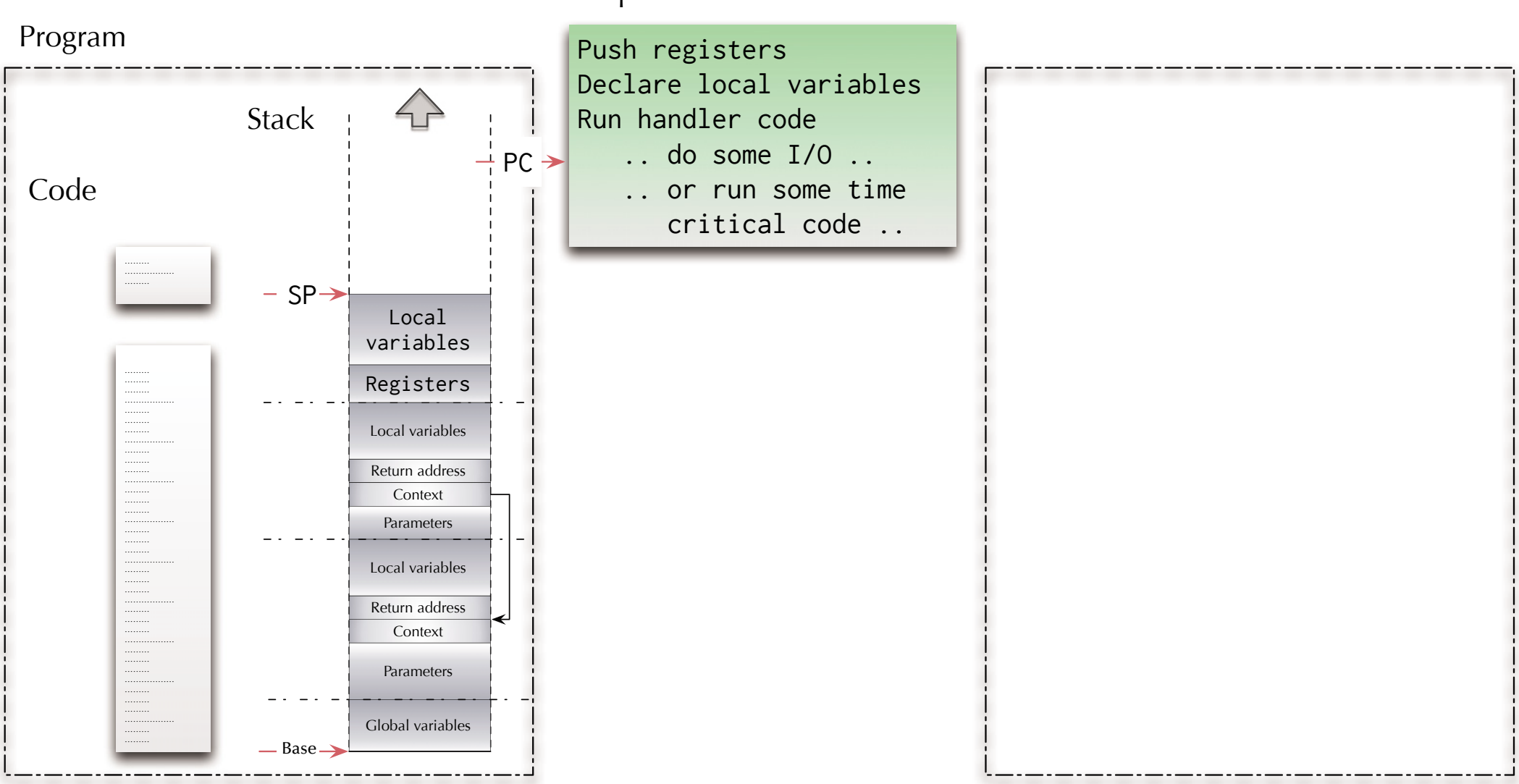

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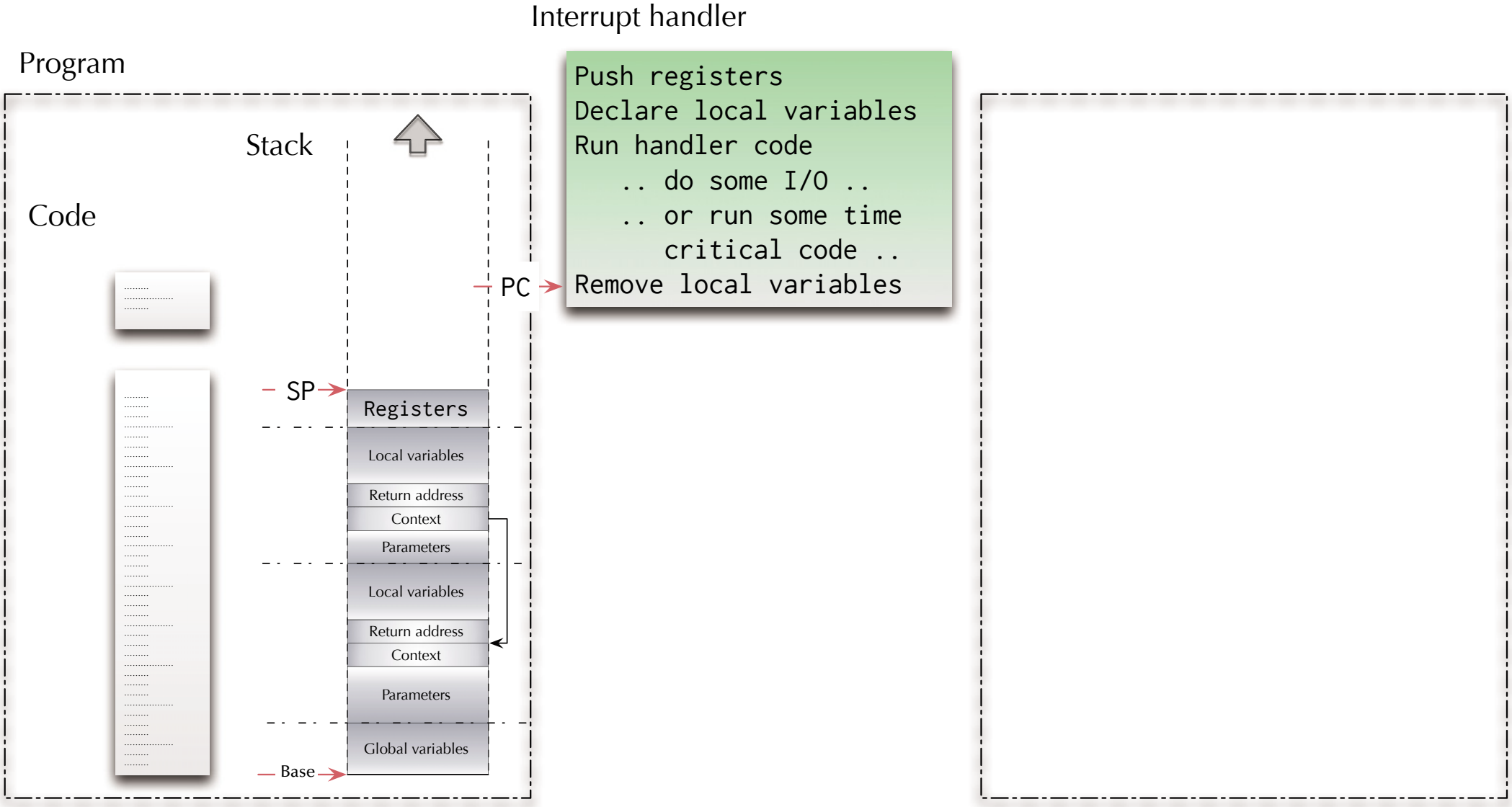

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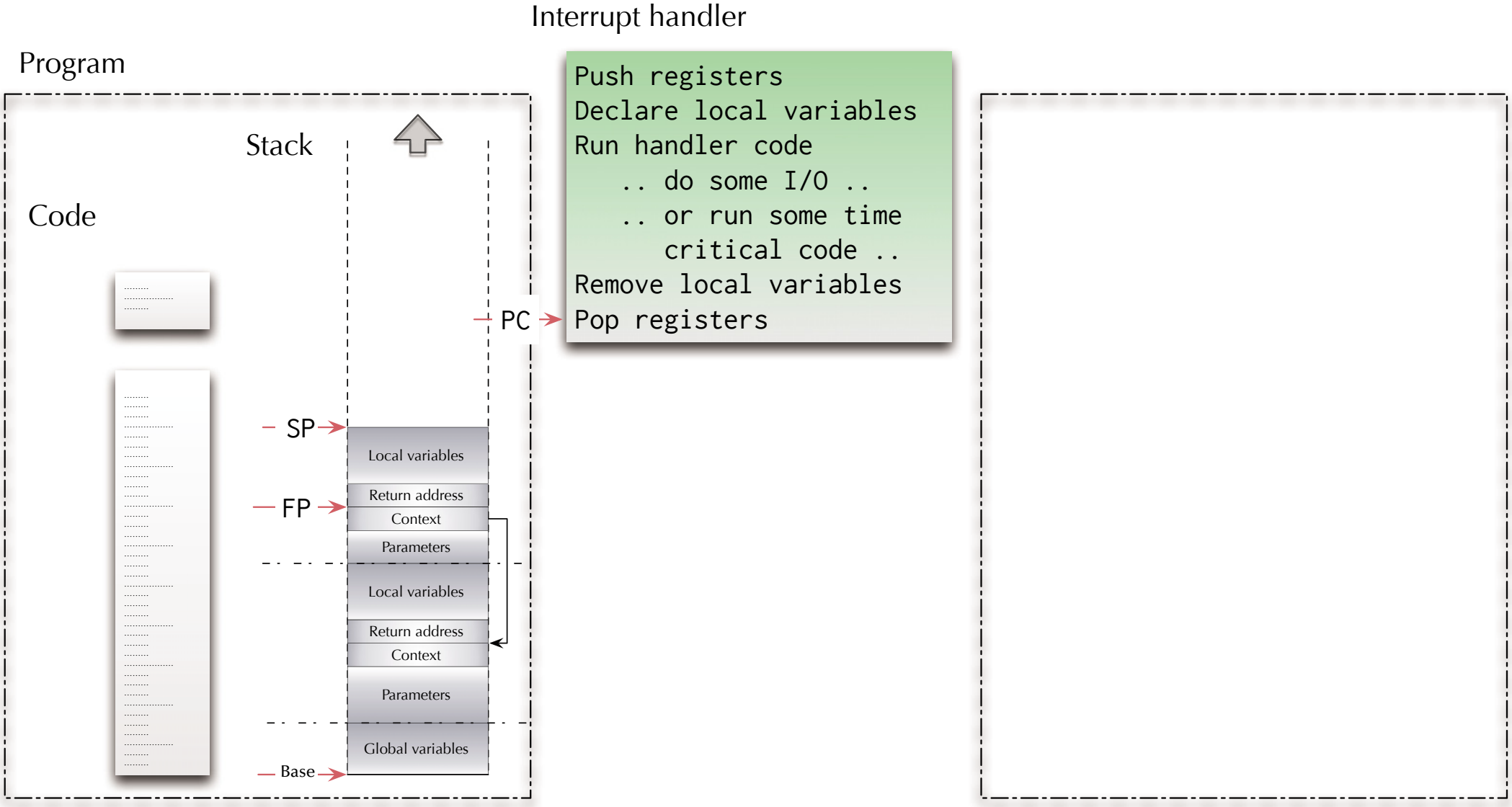

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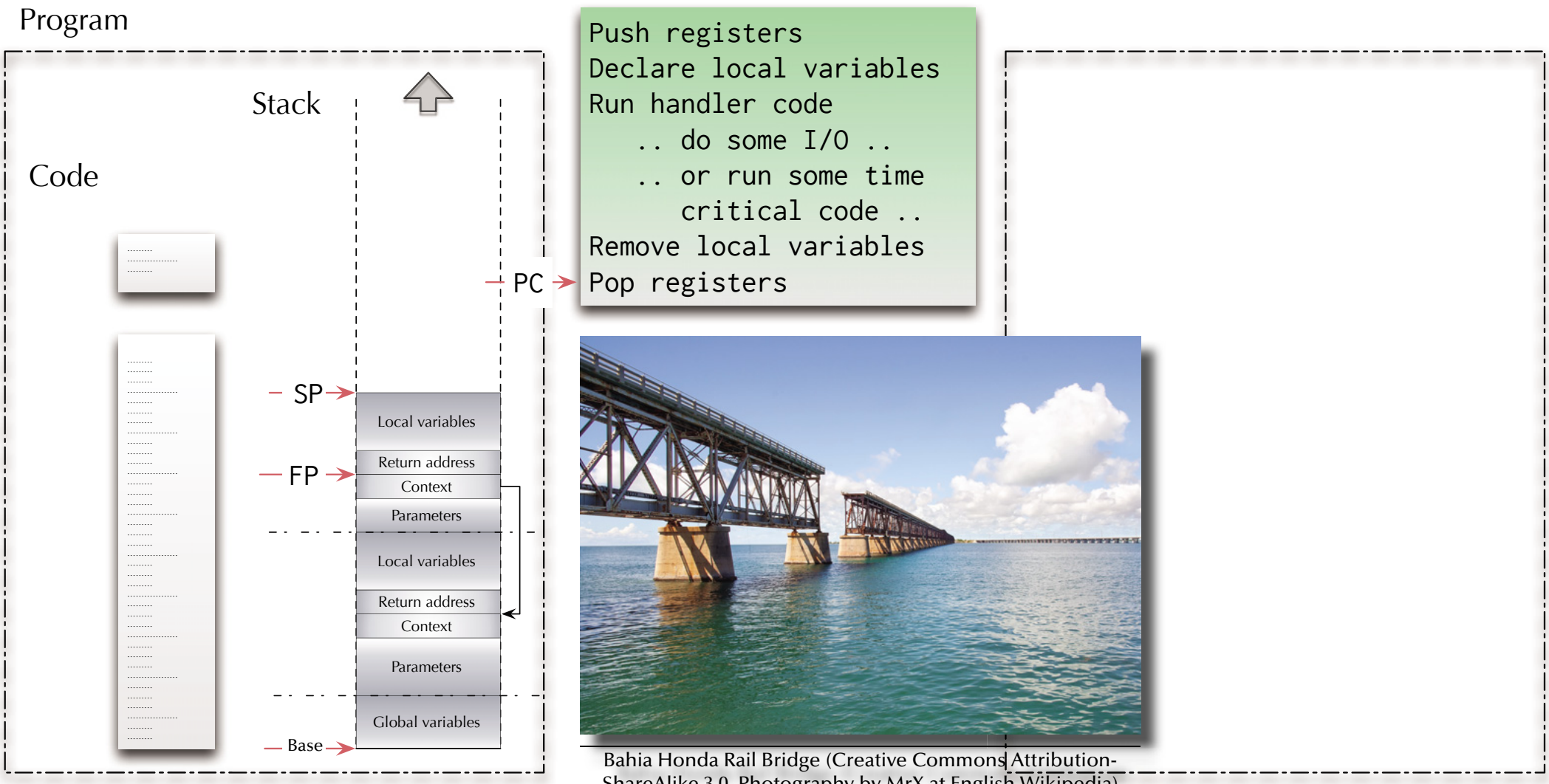

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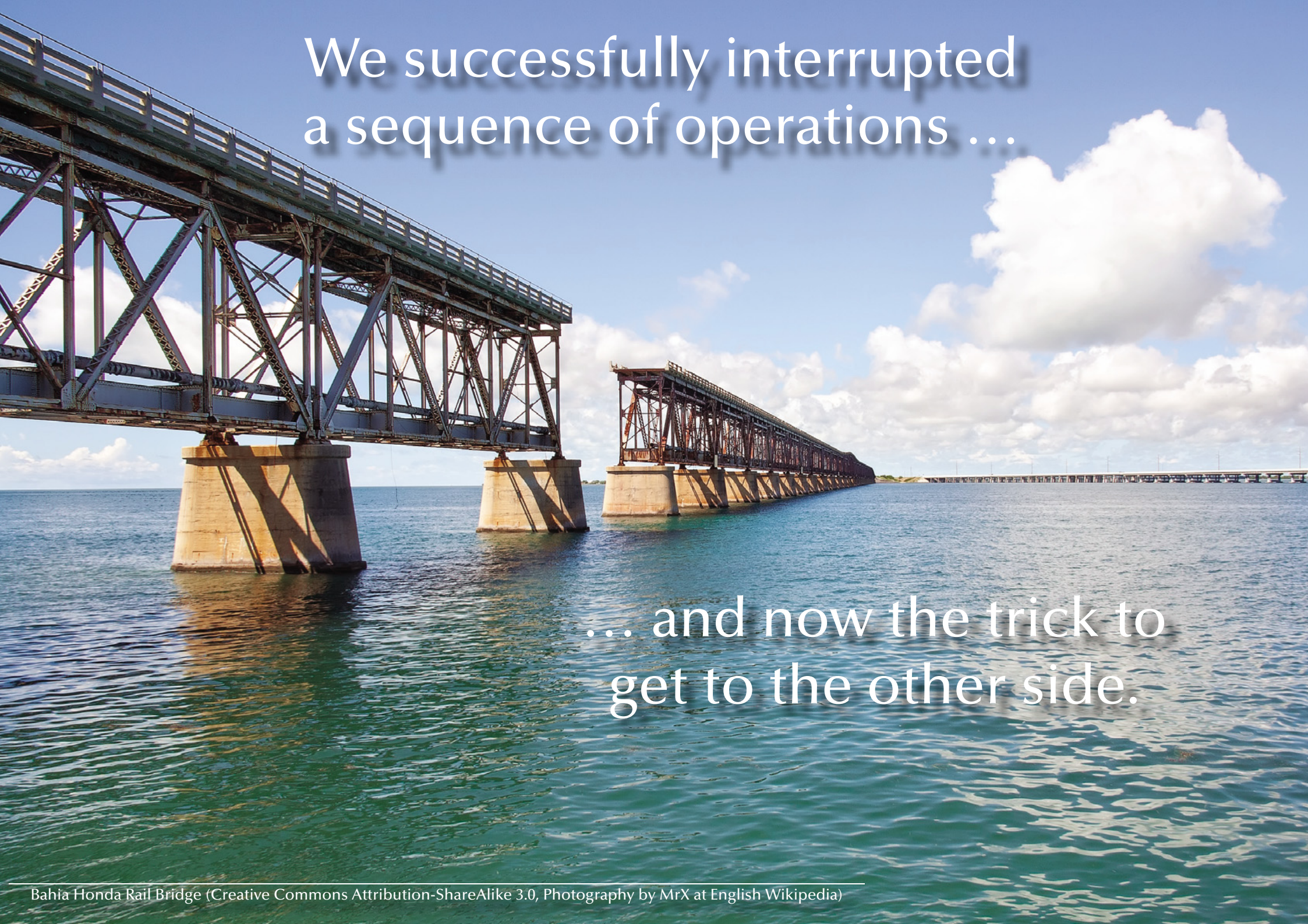

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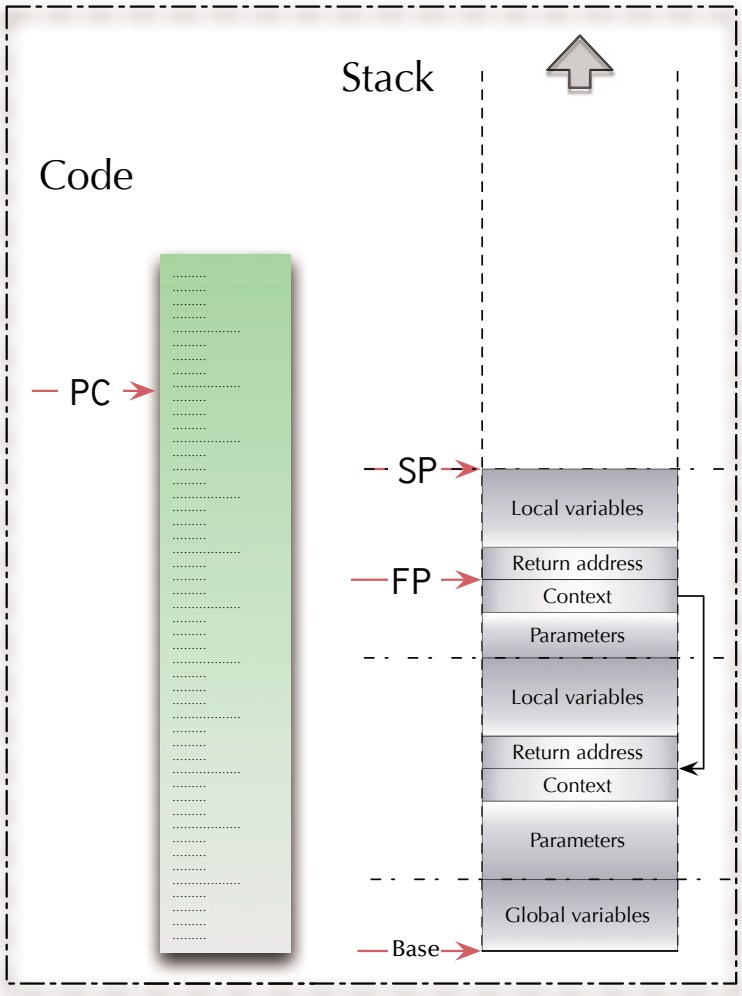
Interrupt processing

Interrupt handler

Program



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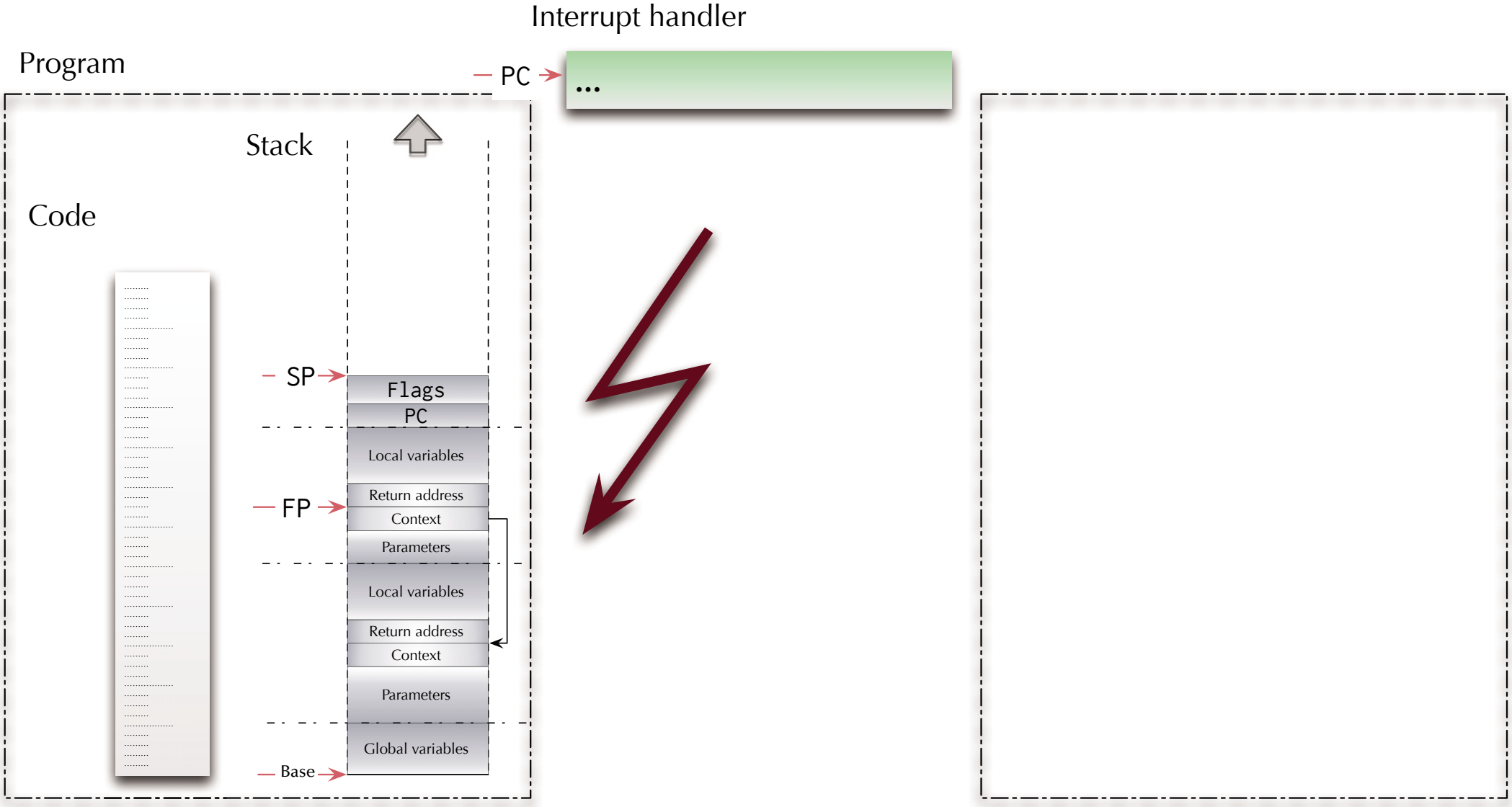

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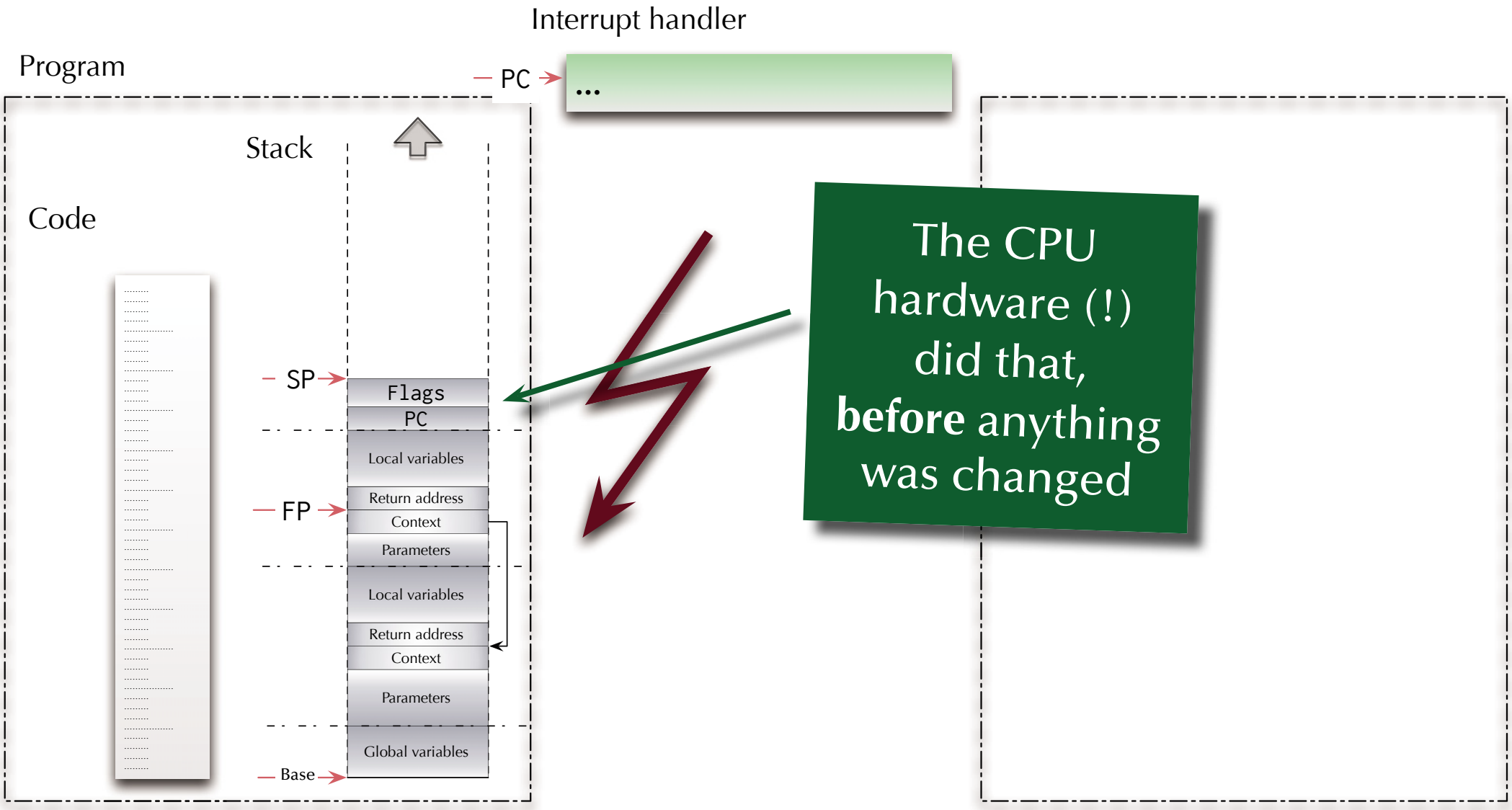

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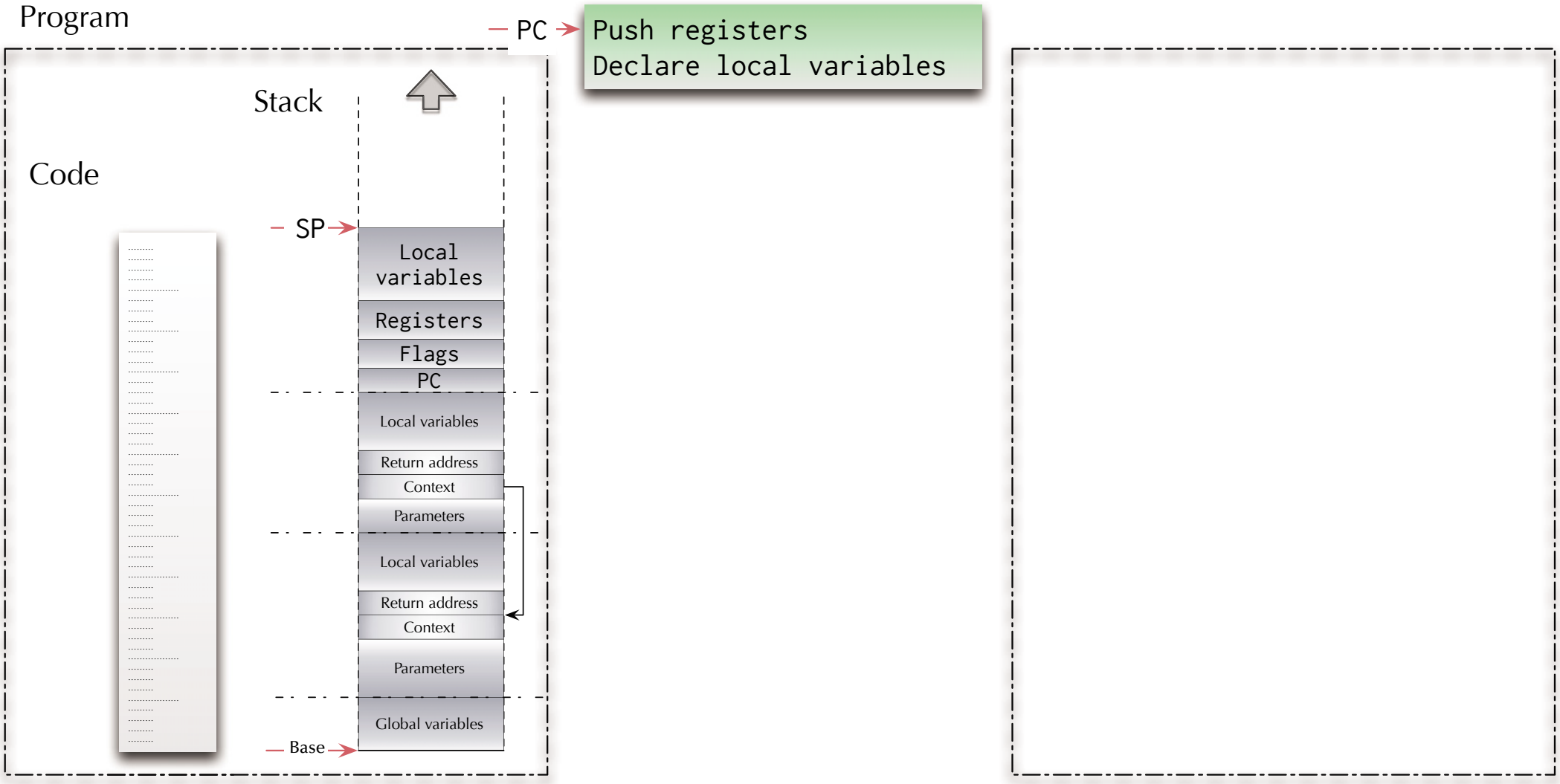

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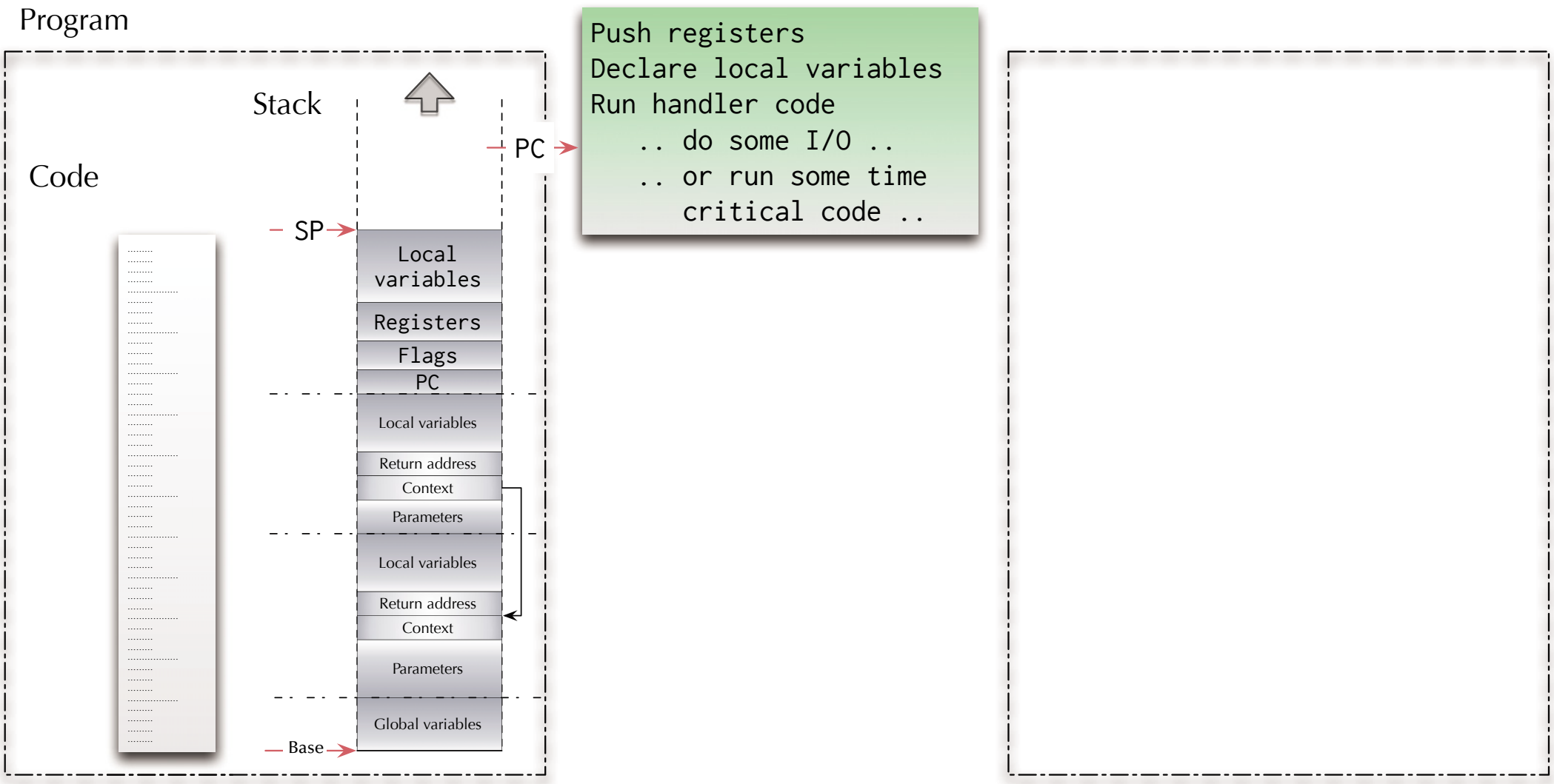

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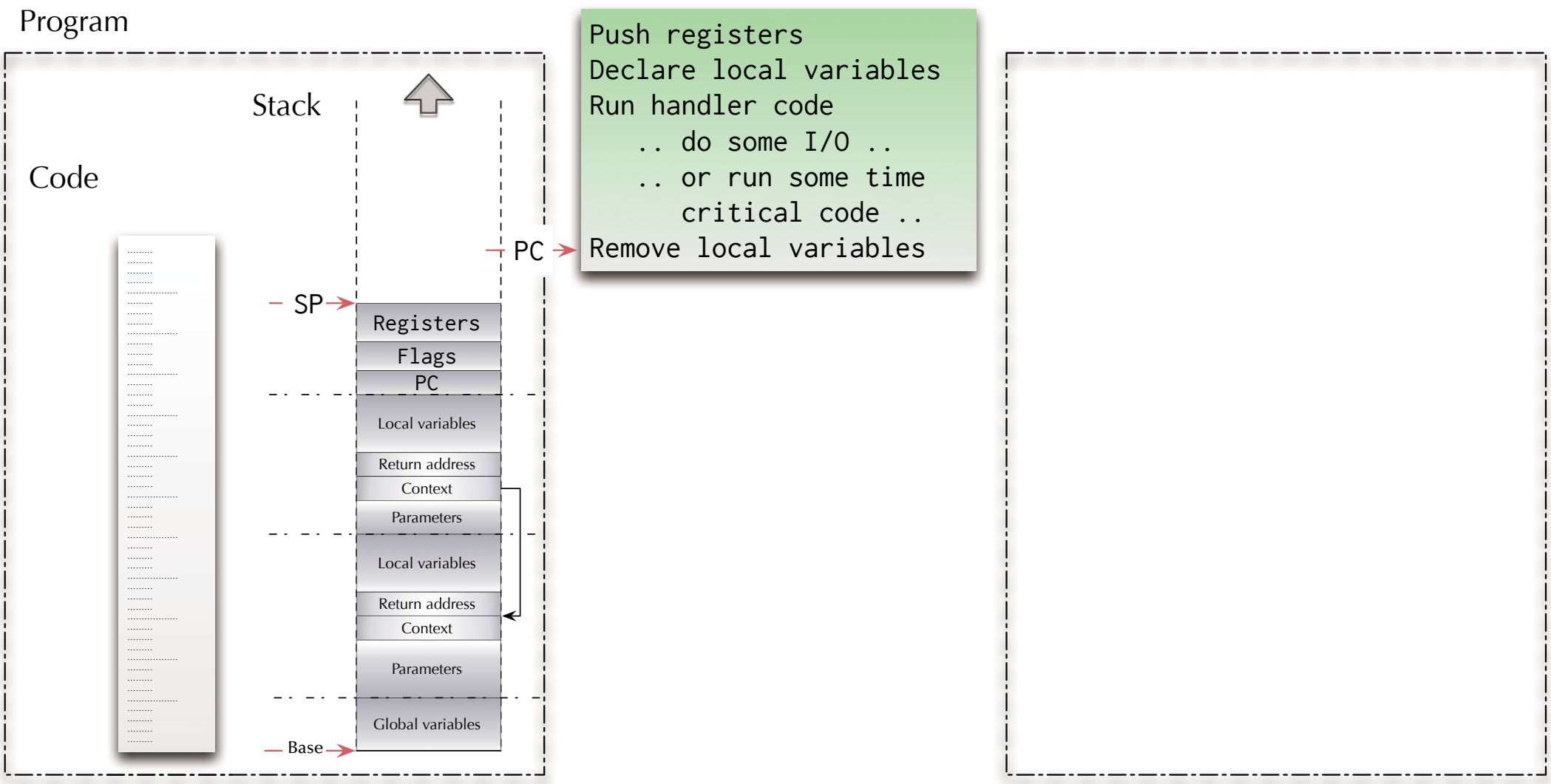

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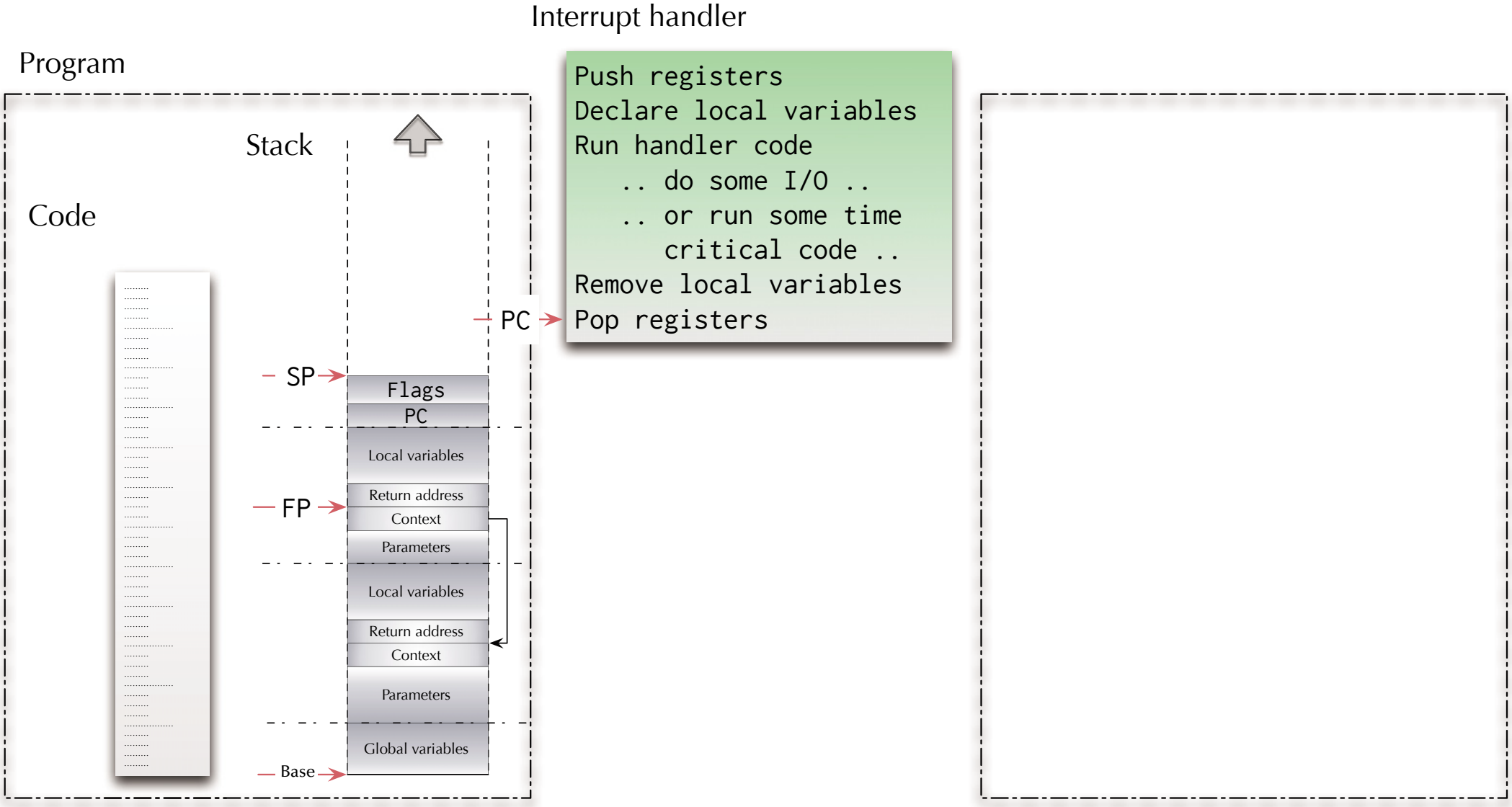
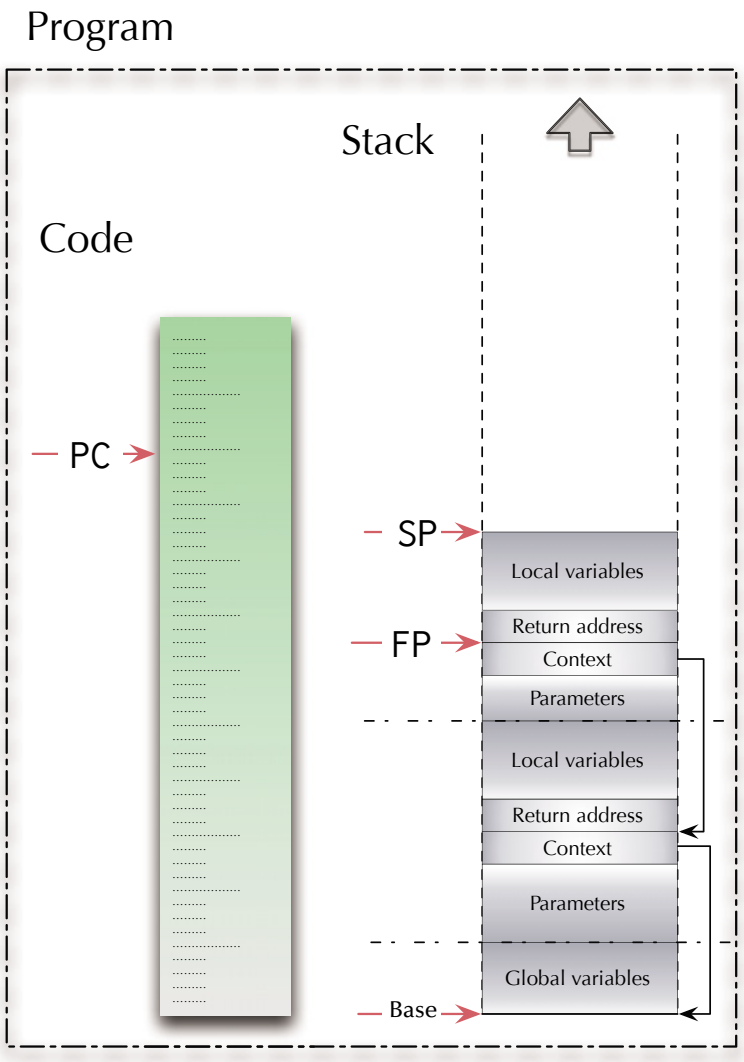


Interrupt processing

Interrupt handler

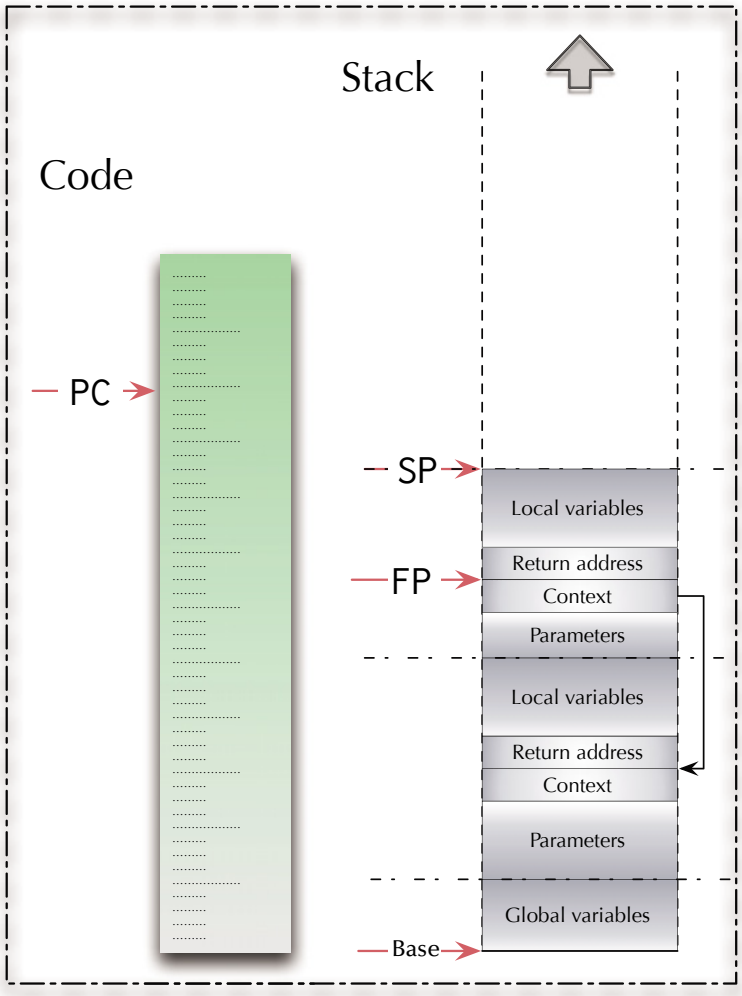



```
Push registers
Declare local variables
Run handler code
   .. do some I/O ..
   .. or run some time
      critical code ..
Remove local variables
Pop registers
Return from interrupt
```

Interrupt processing

Interrupt handler

Program



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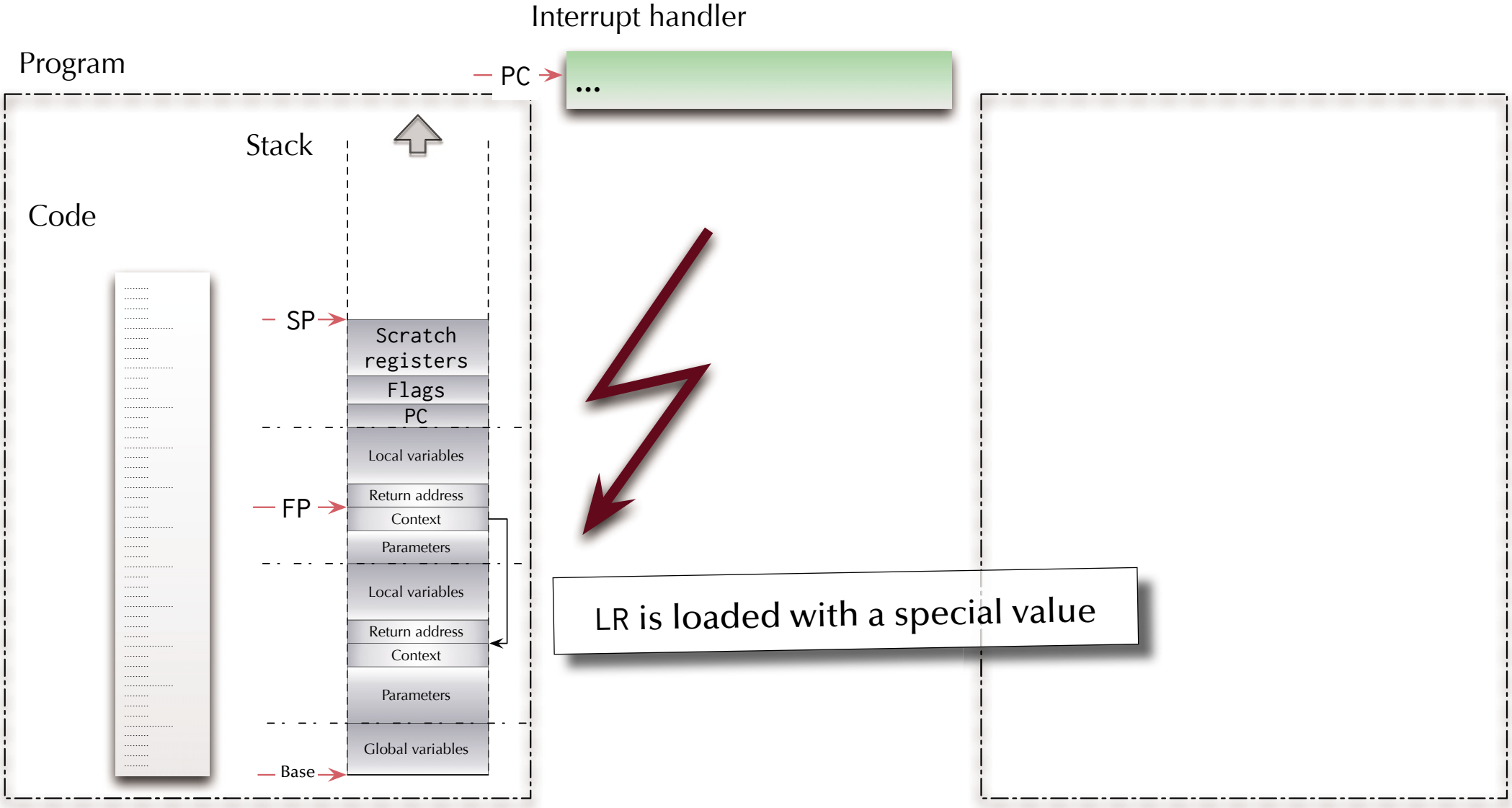

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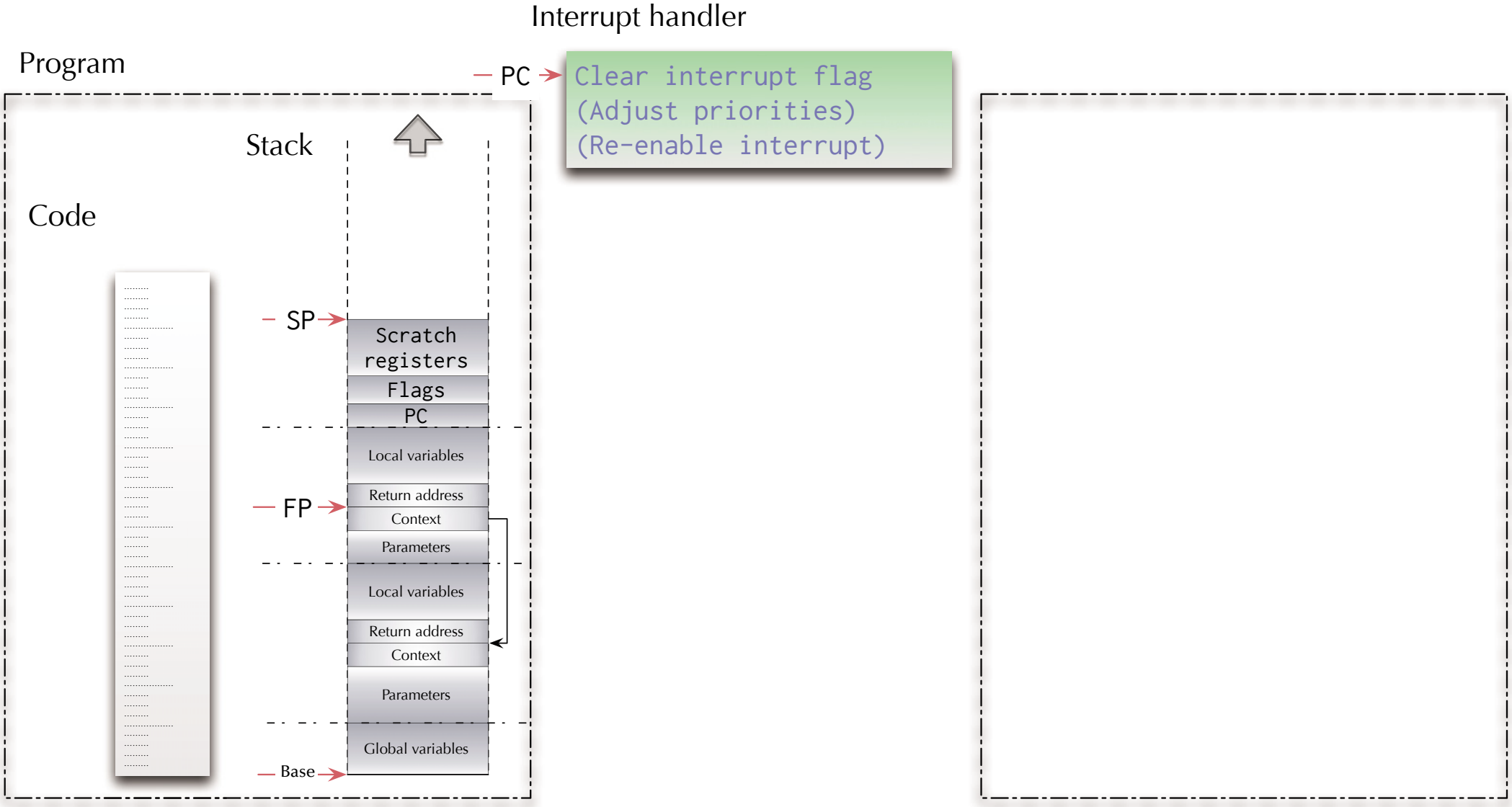

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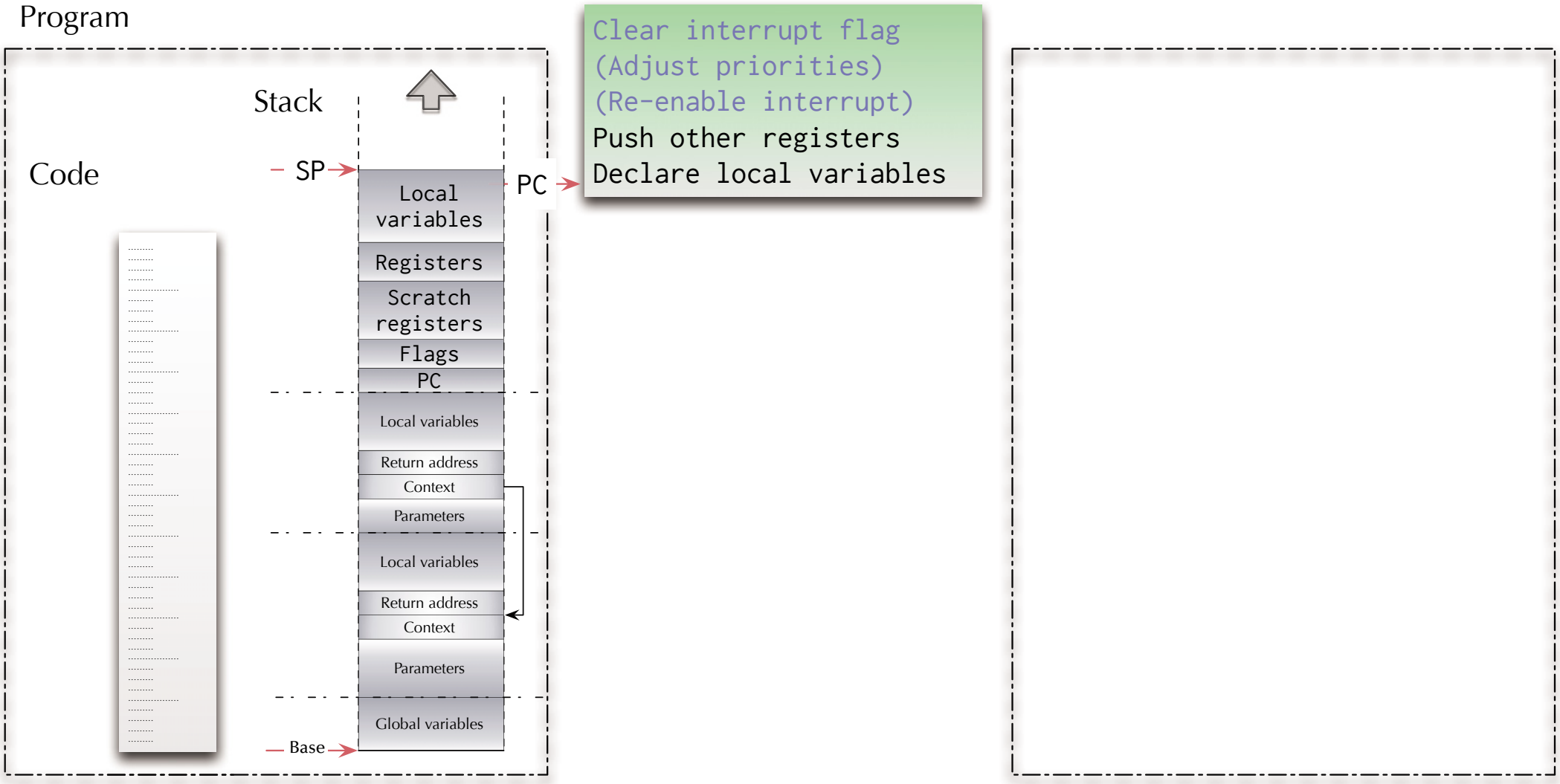

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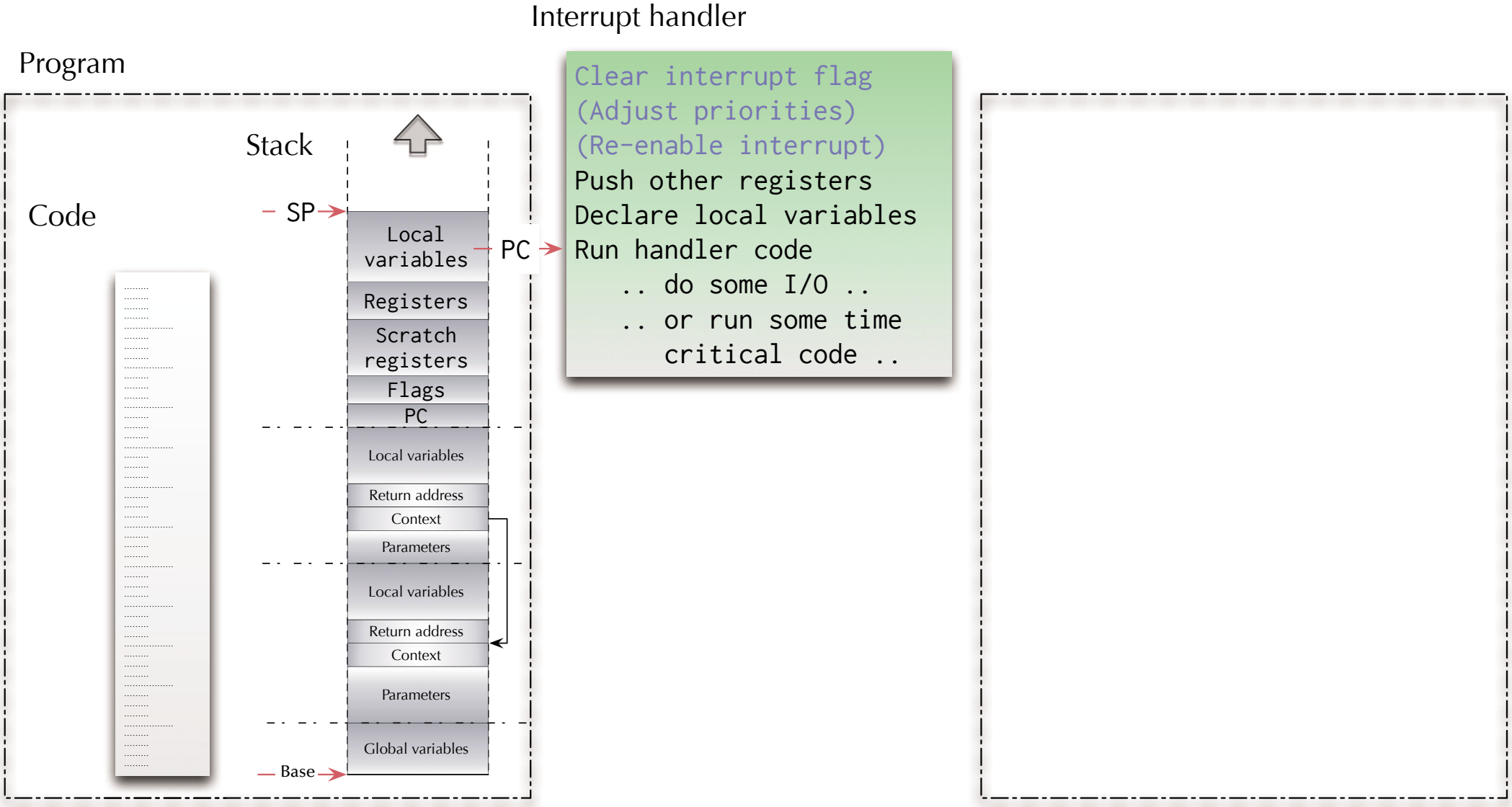

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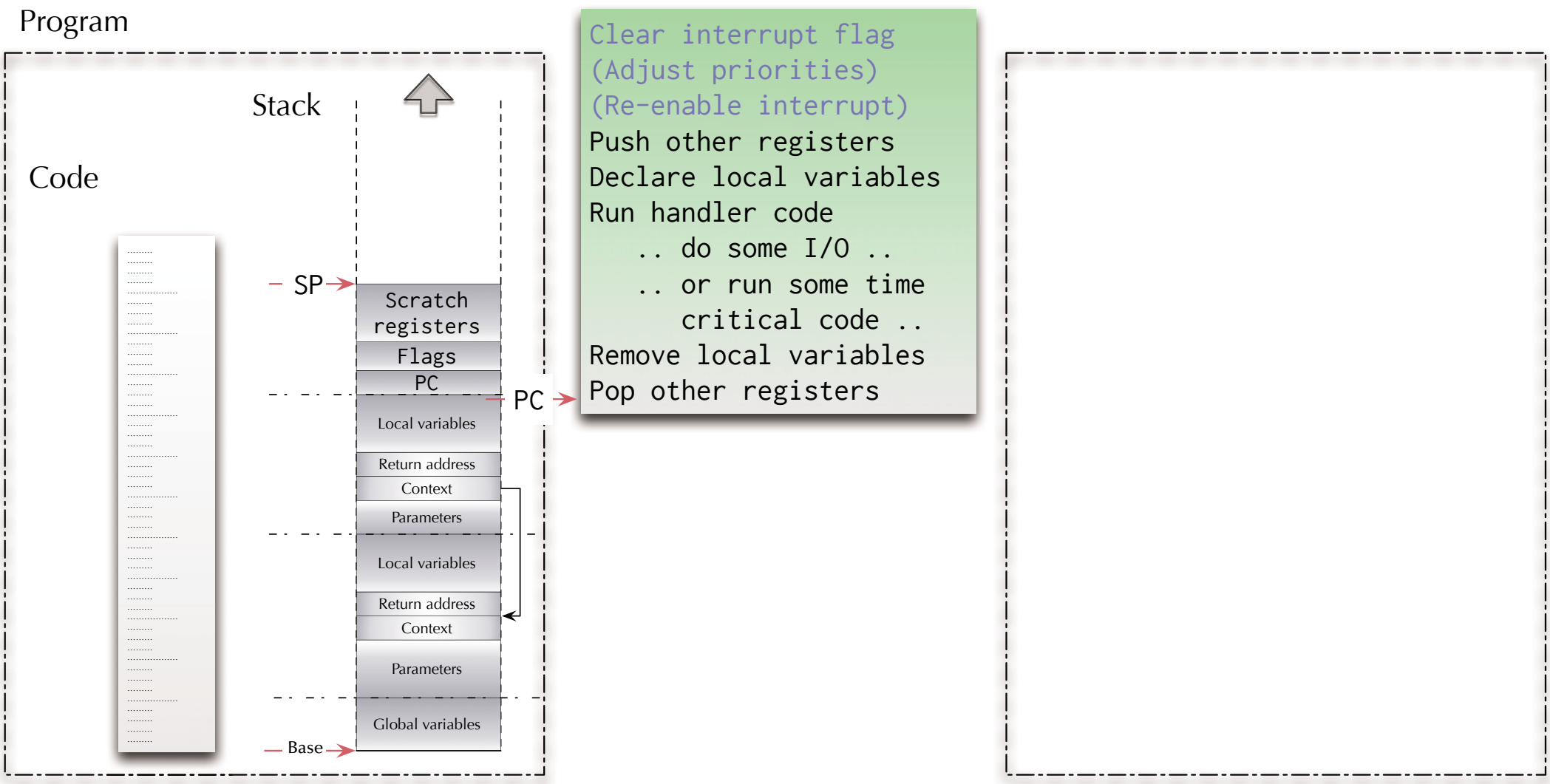

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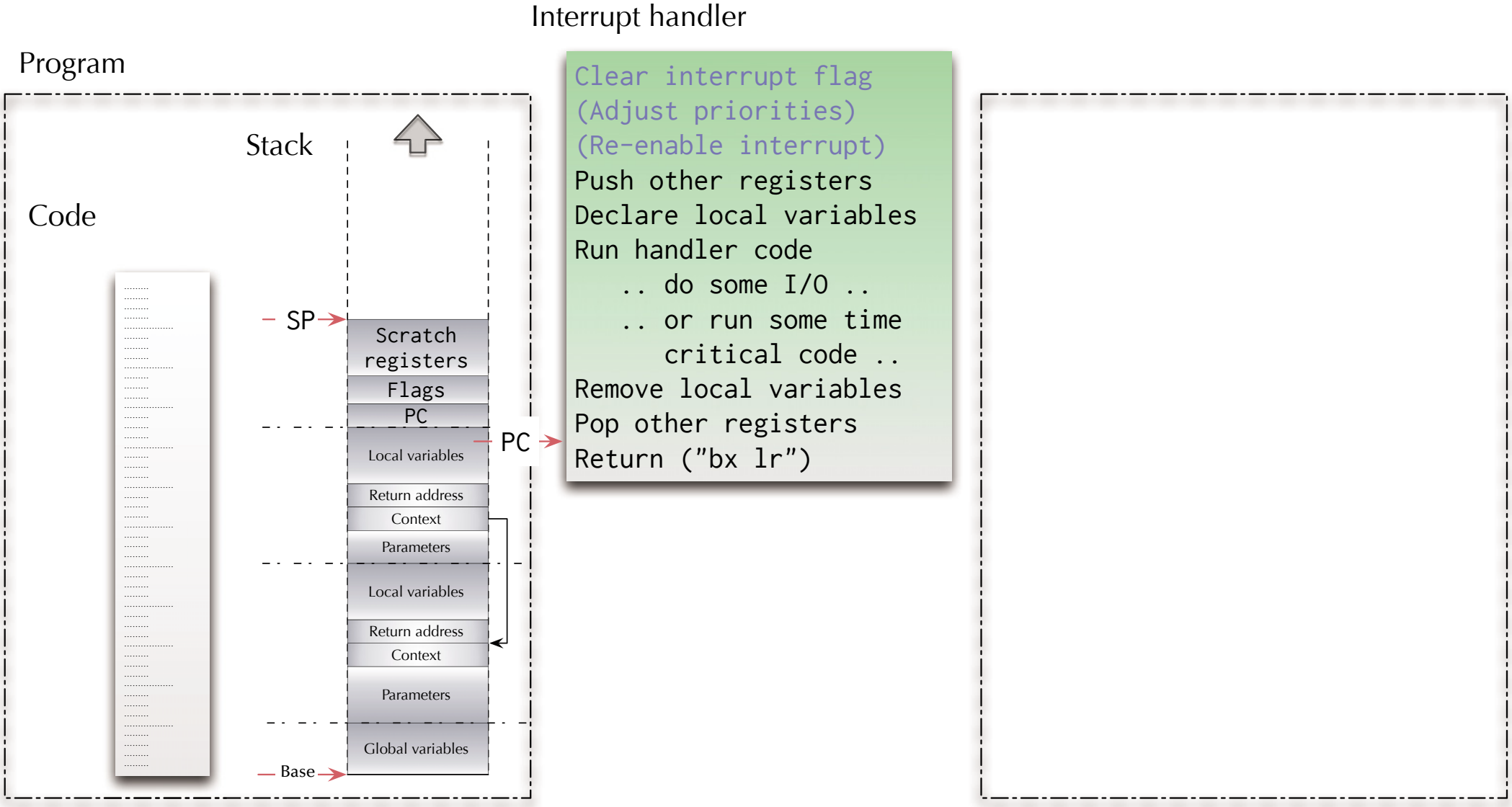
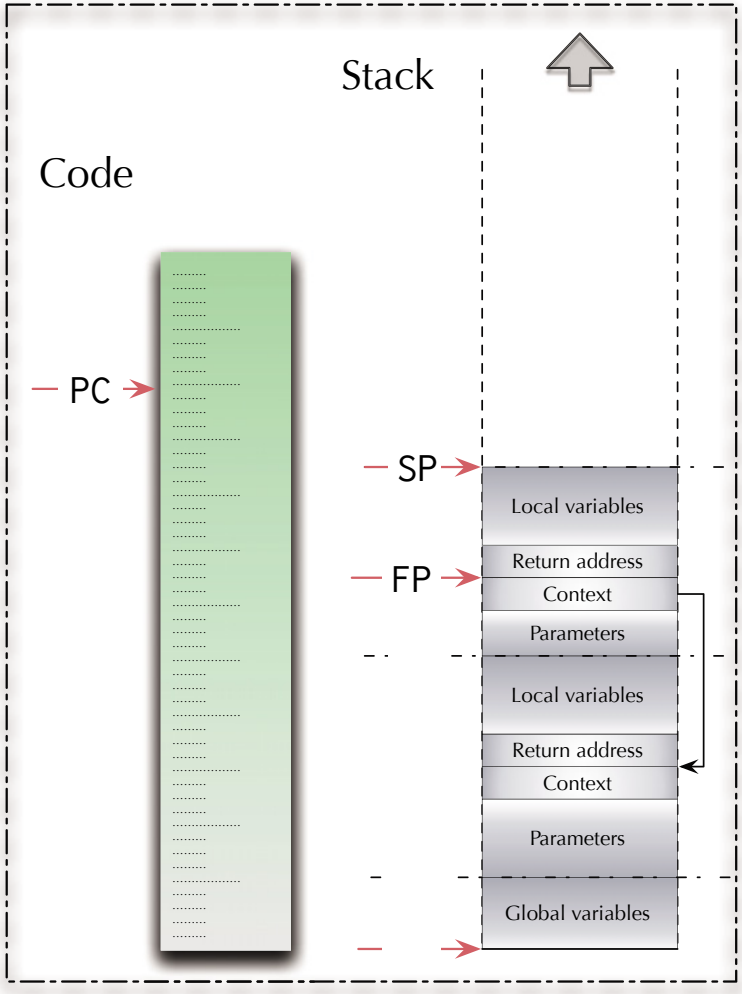


Interrupt processing

Interrupt handler

Program




```
Clear interrupt flag
(Adjust priorities)
(Re-enable interrupt)
Push other registers
Declare local variables
Run handler code
   .. do some I/O ..
   .. or run some time
      critical code ..
Remove local variables
Pop other registers
Return ("bx lr")
```

Interrupt handler

Things to consider

- ☞ Interrupt handler code can be interrupted as well.
- ☞ Are you allowing to interrupt an interrupt handler with an interrupt on the same priority level (e.g. the same interrupt)?
- ☞ Can you overrun a stack with interrupt handlers?

Interrupt handler

Things to consider

- Interrupt handler code can be interrupted as well.
- Are you allowing to interrupt an interrupt handler with an interrupt on the same priority level (e.g. the same interrupt)?
- Can you overrun a stack with interrupt handlers?
- Can we have one of those?

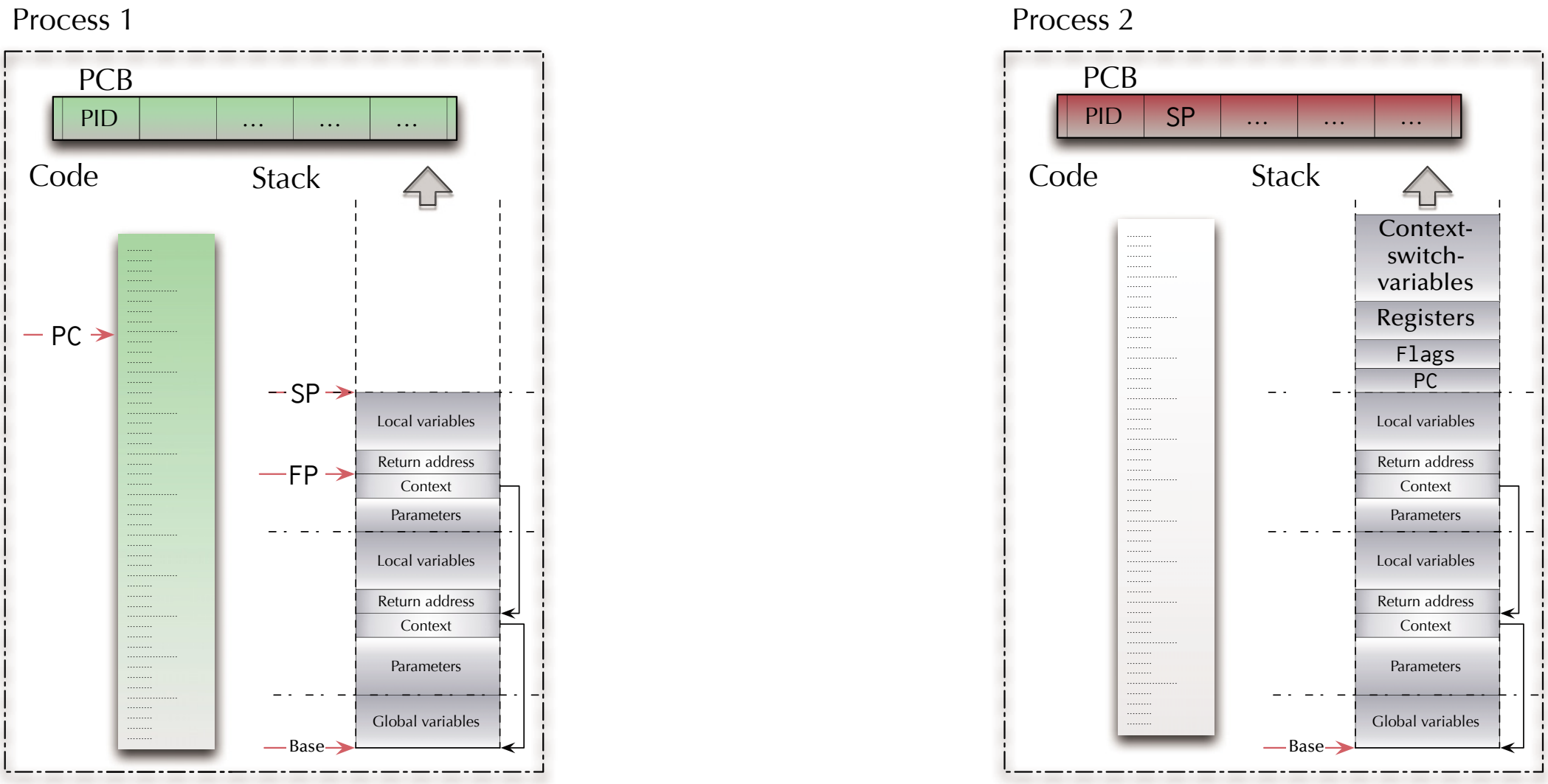
Multiple programs

If we can execute interrupt handler code
“concurrently” to our “main” program:

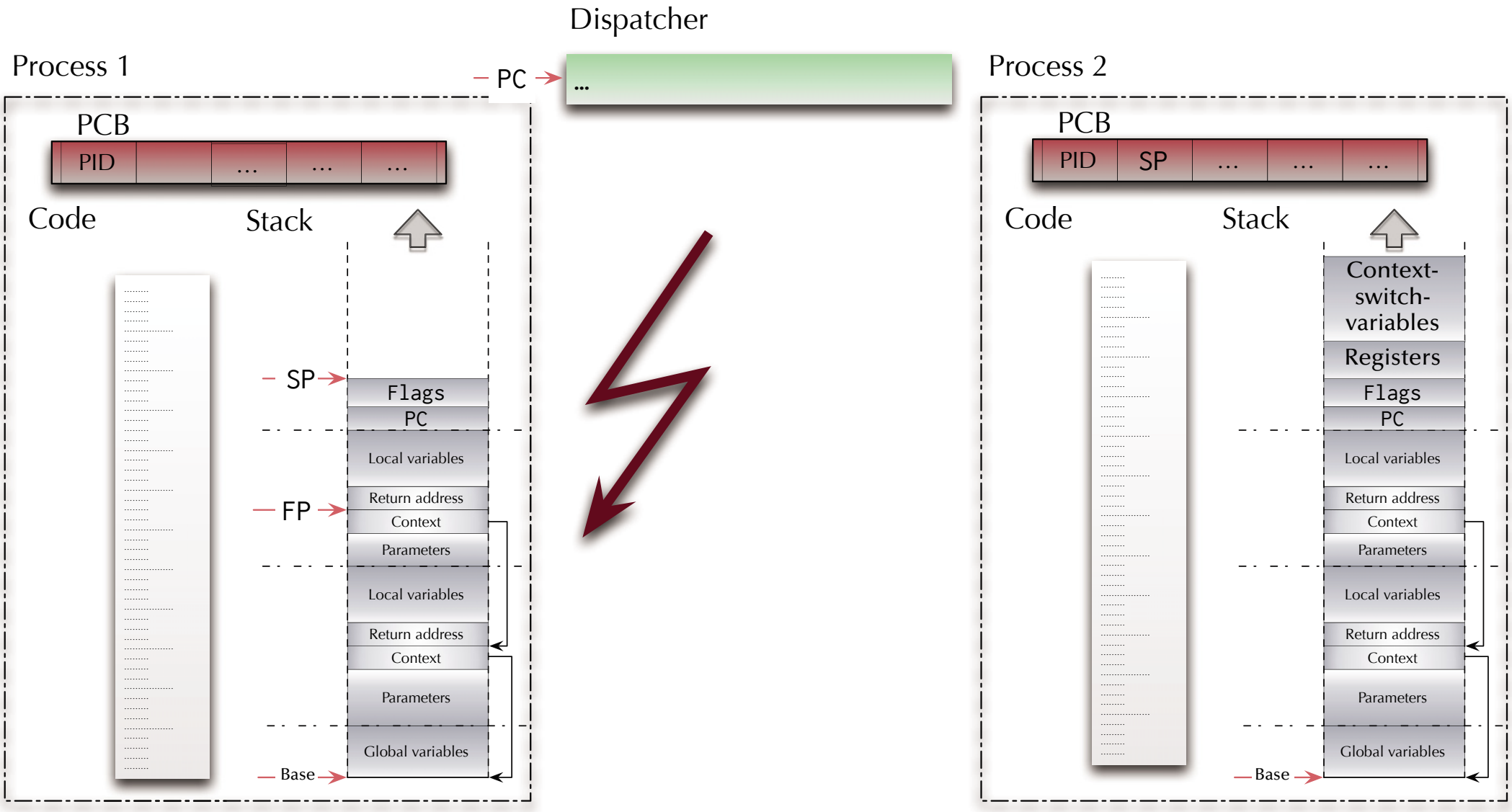
☞ Can we then also have multiple “main” programs?

Context switch

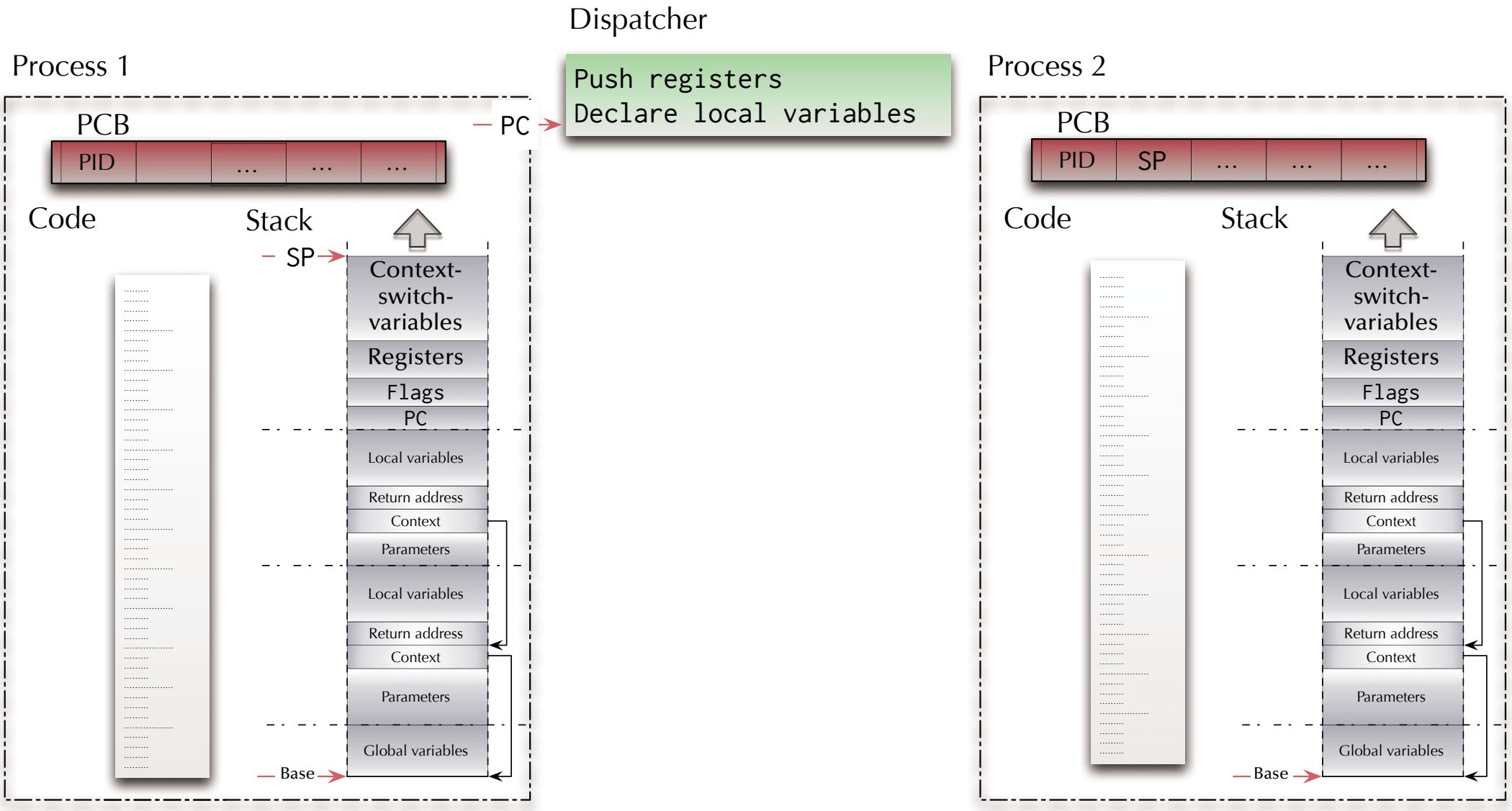
Dispatcher



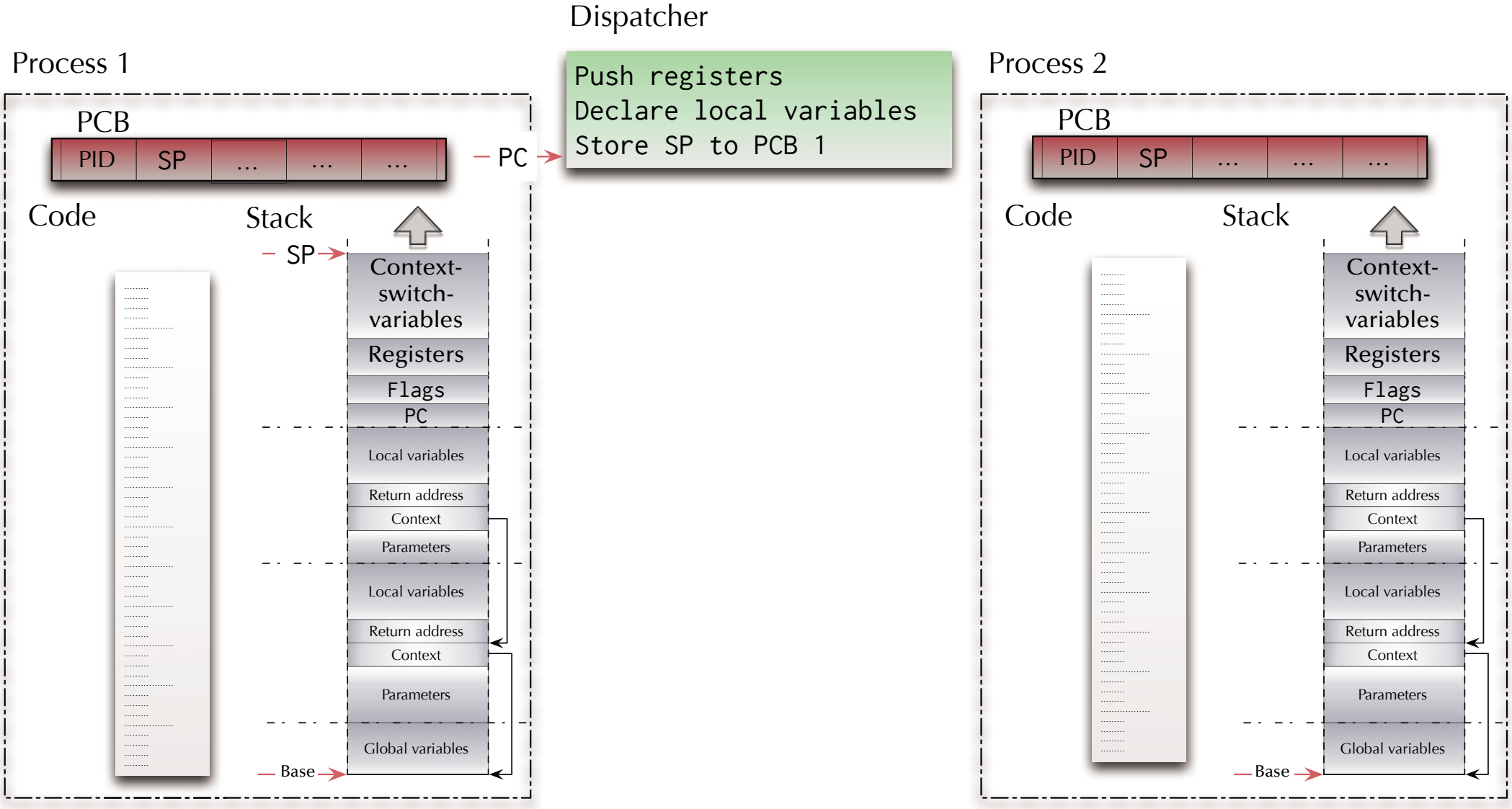
Context switch



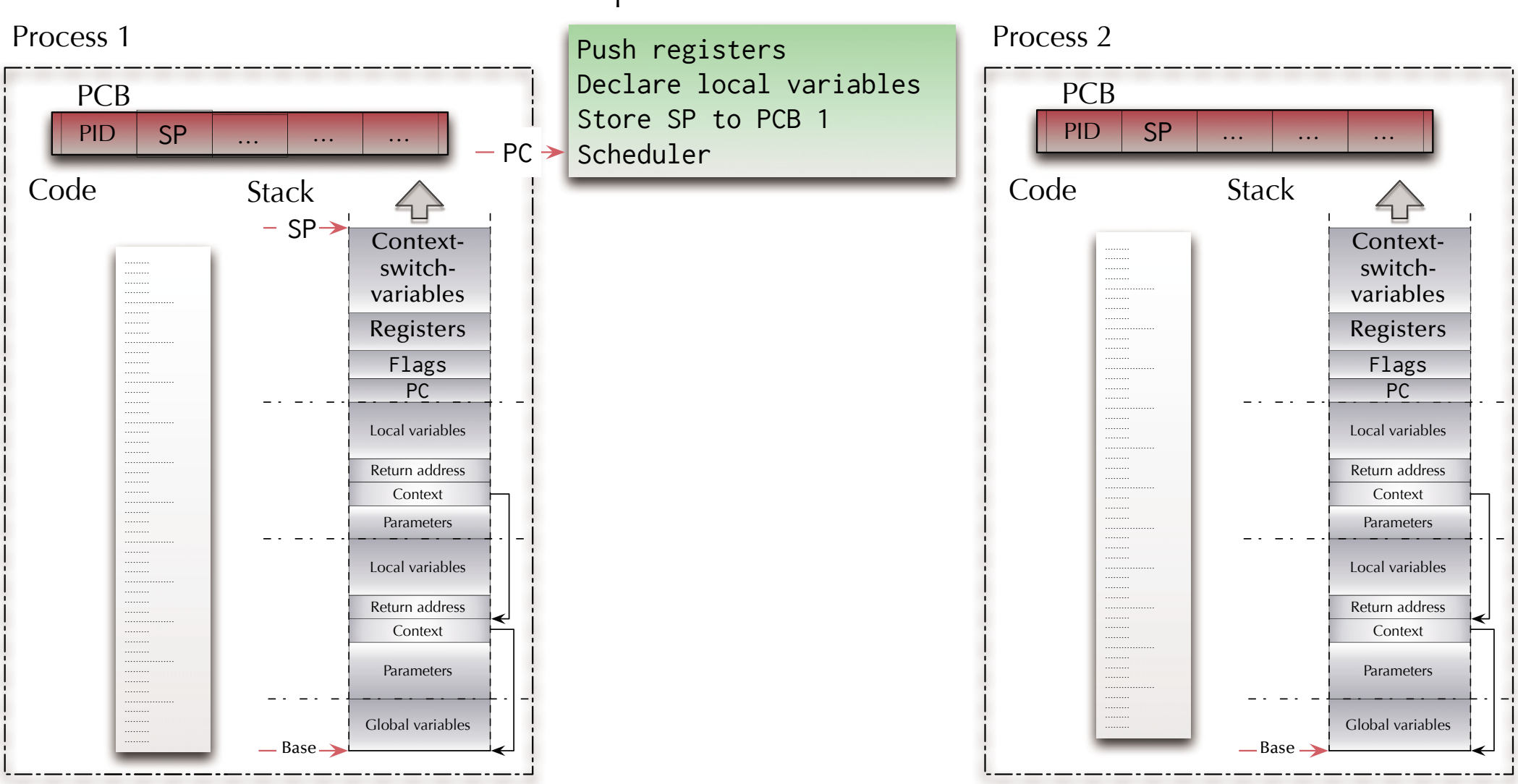
Context switch



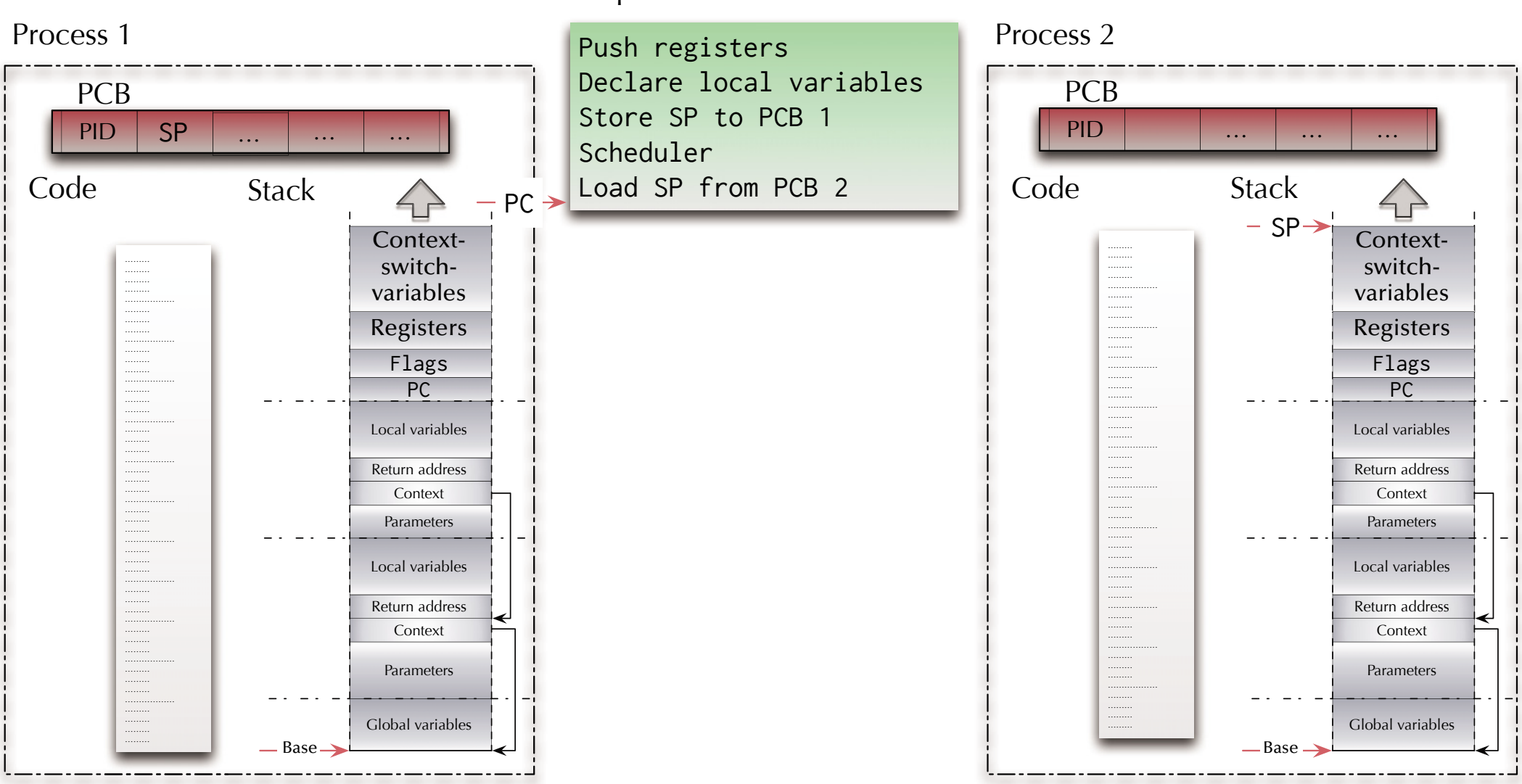
Context switch



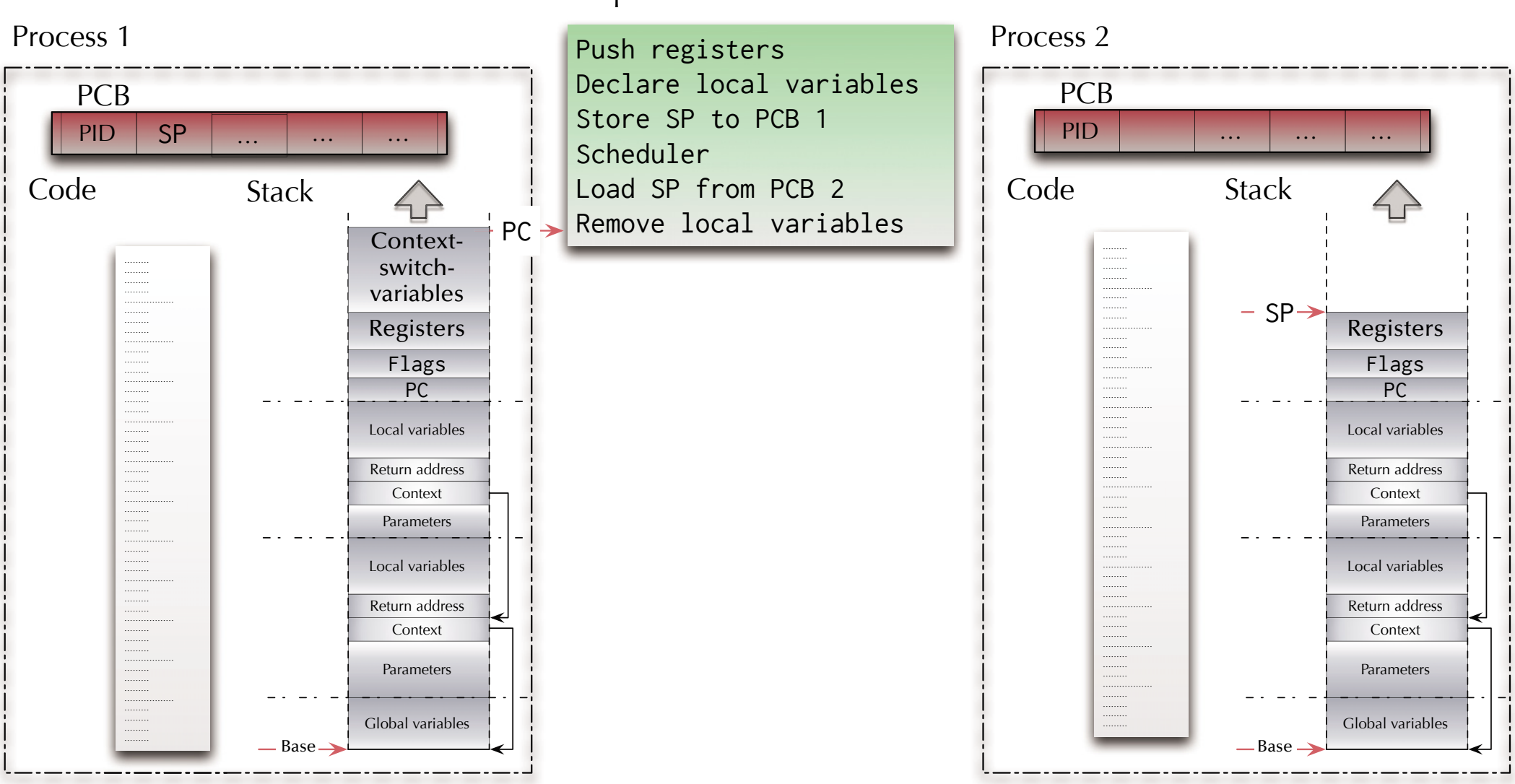
Context switch



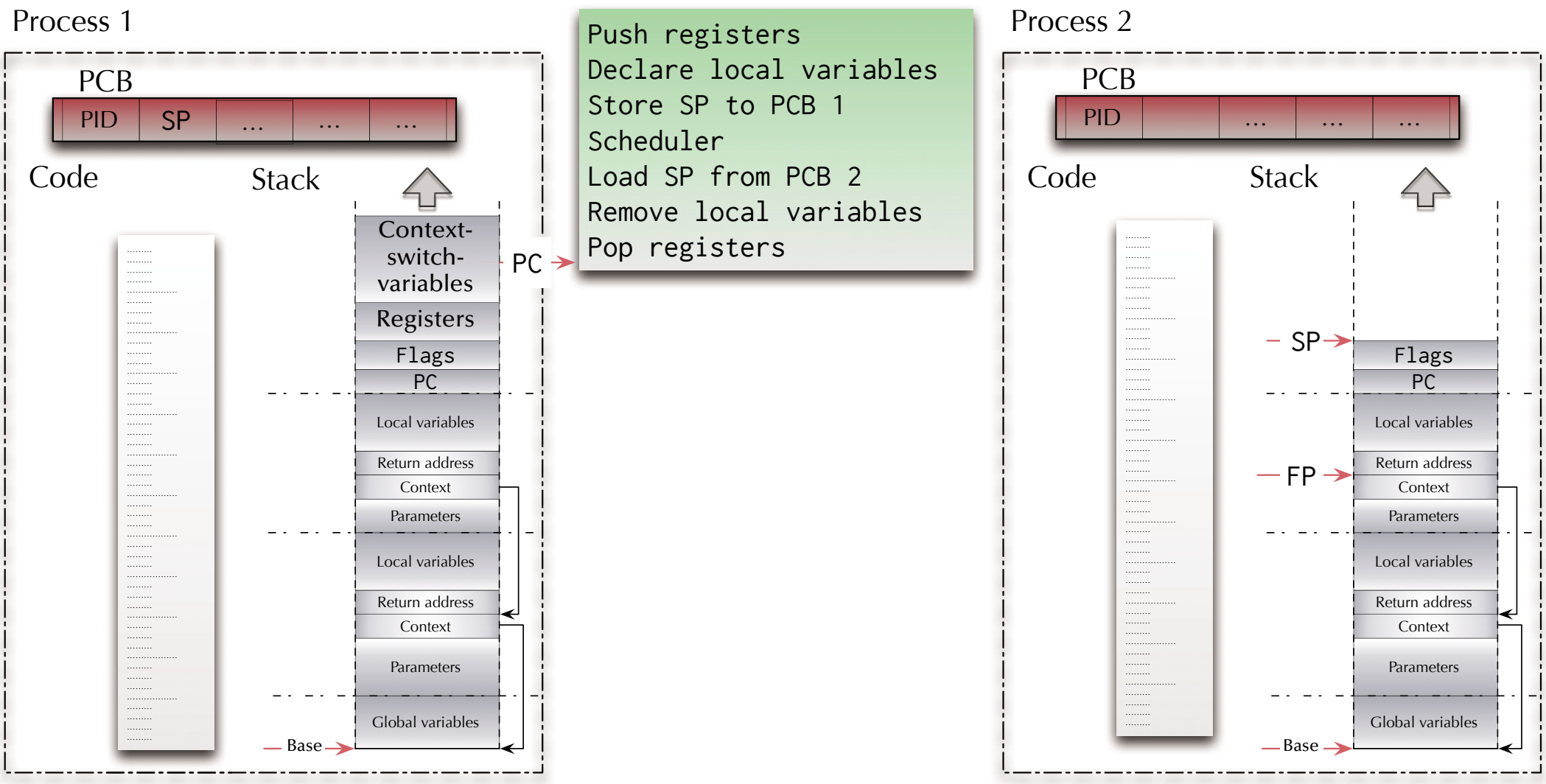
Context switch



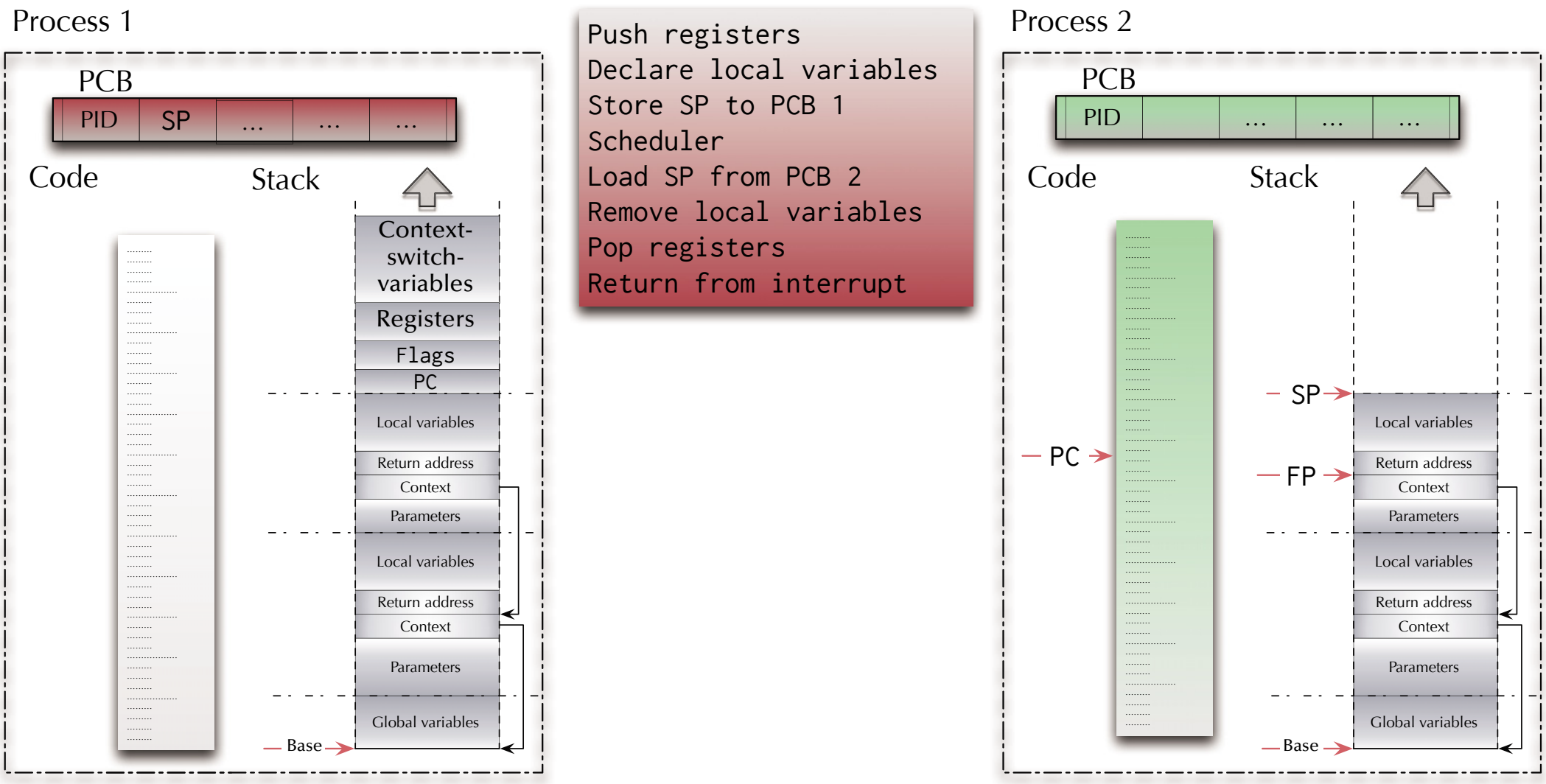
Context switch



Context switch



Context switch



Multi-tasking and Contention

Anything else could go wrong?

Multi-tasking and Contention

Anything else could go wrong?

☞ If there is neither communication nor contention between concurrent parts
… all is easy … and boring.

☞ What happens if concurrent programs **share** data?

Shared variables

Atomic load & store operations

☞ Assumption 1: every individual base memory cell (word) load and store access is *atomic*

☞ Assumption 2: there is *no* atomic combined load-store access

```
G : Natural := 0; -- assumed to be mapped on a 1-word cell in memory
```

```
task body P1 is
begin
   G := 1
   G := G + G;
end P1;
```

```
task body P2 is
begin
   G := 2
   G := G + G;
end P2;
```

```
task body P3 is
begin
   G := 3
   G := G + G;
end P3;
```

☞ What is the value of `G`?

Shared variables

Atomic load & store operations

☞ Assumption 1: every individual base memory cell (word) load and store access is *atomic*

☞ Assumption 2: there is *no* atomic combined load-store access

```
G: .word 0x00000000
```

```
ldr   r4, =G
mov   r1, #1
str   r1, [r4]
ldr   r2, [r4]
ldr   r3, [r4]
add   r1, r2, r3
str   r1, [r4]
```

```
ldr   r4, =G
mov   r1, #2
str   r1, [r4]
ldr   r2, [r4]
ldr   r3, [r4]
add   r1, r2, r3
str   r1, [r4]
```

```
ldr   r4, =G
mov   r1, #3
str   r1, [r4]
ldr   r2, [r4]
ldr   r3, [r4]
add   r1, r2, r3
str   r1, [r4]
```

☞ What is the value in memory cell `G` after all three programs complete?

Shared variables

This is terrible!

Nobody is their right mind would analyse a program like that.

☞ … are we missing something?

☞ … is there an elegant way out?

Mutual exclusion … or the lack thereof

```
Count : Integer := 0;
```

```
task body Enter is
begin
 for i := 1 .. 100 loop
  Count := Count + 1;
 end loop;
end Enter;
```

```
task body Leave is
begin
 for i := 1 .. 100 loop
  Count := Count - 1;
 end loop;
end Leave;
```

☞ What is the value of `Count` after both programs complete?

Mutual exclusion … or the lack thereof

```
Count: .word 0x00000000
```

```
   ldr   r4, =Count
   mov   r1, #1
for_enter:
   cmp   r1, #100
   bgt   end_for_enter
   ldr   r2, [r4]
   add   r2, #1
   str   r2, [r4]
   add   r1, #1
   b     for_enter
end_for_enter:
```

```
   ldr   r4, =Count
   mov   r1, #1
for_leave:
   cmp   r1, #100
   bgt   end_for_leave
   ldr   r2, [r4]
   sub   r2, #1
   str   r2, [r4]
   add   r1, #1
   b     for_leave
end_for_leave:
```

☞ What is the value at address Count after both programs complete?

Mutual exclusion … or the lack thereof

```
Count: .word 0x00000000
```

```
    ldr   r4, =Count
    mov   r1, #1
for_enter:
    cmp   r1, #100
    bgt   end_for_enter
    ldr   r2, [r4]
    add   r2, #1
    str   r2, [r4]
    add   r1, #1
    b     for_enter
end_for_enter:
```

```
    ldr   r4, =Count
    mov   r1, #1
for_leave:
    cmp   r1, #100
    bgt   end_for_leave
    ldr   r2, [r4]
    sub   r2, #1
    str   r2, [r4]
    add   r1, #1
    b     for_leave
end_for_leave:
```

☞ What is the value at address Count after both programs complete?

Mutual exclusion … or the lack thereof

```
Count: .word 0x00000000
```

```
   ldr   r4, =Count
   mov   r1, #1
for_enter:
   cmp   r1, #100
   bgt   end_for_enter
enter_strex_fail:
   ldrex r2, [r4] ; tag [r4] as exclusive
   add   r2, #1
   strex r2, [r4] ; only if untouched
   cbnz  r2, enter_strex_fail
   add   r1, #1
   b     for_enter
end_for_enter:
```

```
   ldr   r4, =Count
   mov   r1, #1
for_leave:
   cmp   r1, #100
   bgt   end_for_leave
leave_strex_fail:
   ldrex r2, [r4] ; tag [r4] as exclusive
   sub   r2, #1
   strex r2, [r4] ; only if untouched
   cbnz  r2, leave_strex_fail
   add   r1, #1
   b     for_leave
end_for_leave:
```

☞ What is the value at address Count after both programs complete?

Mutual exclusion

```
Count: .word 0x00000000
```

```
    ldr   r4, =Count
    mov   r1, #1
for_enter:
    cmp   r1, #100
    bgt   end_for_enter
enter_strex_fail:
    ldrex r2, [r4] ; tag [r4] as exclusive
    add   r2, #1
    strex r2, [r4] ; only if untouched
    cbnz  r2, enter_strex_fail
    add   r1, #1
    b     for_enter
end_for_enter:
```

```
    ldr   r4, =Count
    mov   r1, #1
for_leave:
    cmp   r1, #100
    bgt   end_for_leave
leave_strex_fail:
    ldrex r2, [r4] ; tag [r4] as exclusive
    sub   r2, #1
    strex r2, [r4] ; only if untouched
    cbnz  r2, leave_strex_fail
    add   r1, #1
    b     for_leave
end_for_leave:
```

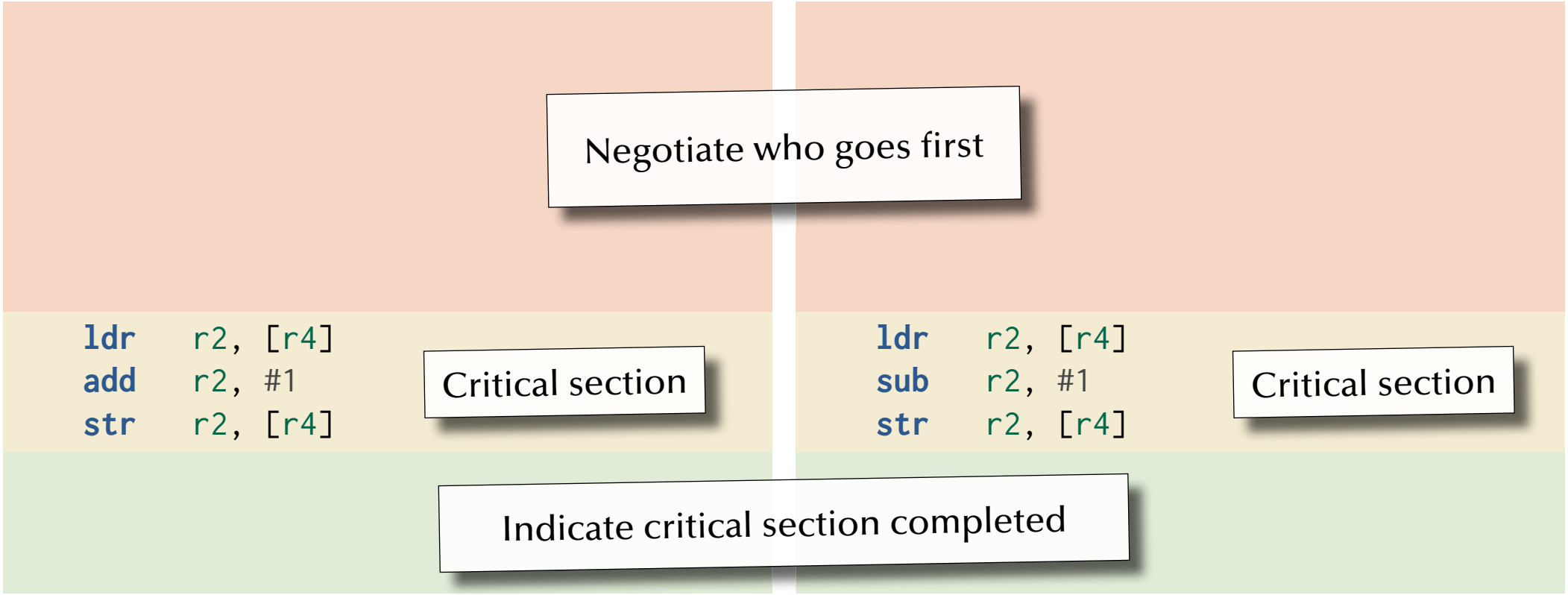
Any context switch needs to clear reservations

☞ What is the value at address Count after both programs complete?

```
Count:    .word 0x00000000
```

```
    ldr   r4, =Count
    mov   r1, #1
for_enter:
    cmp   r1, #100
    bgt   end_for_enter
```

```
    ldr   r4, =Count
    mov   r1, #1
for_leave:
    cmp   r1, #100
    bgt   end_for_leave
```




```
    ldr   r2, [r4]
    add   r2, #1
    str   r2, [r4]
```

```
    ldr   r2, [r4]
    sub   r2, #1
    str   r2, [r4]
```

```
    add   r1, #1
    b     for_enter
end_for_enter:
```

```
    add   r1, #1
    b     for_leave
end_for_leave:
```

```
Count:    .word 0x00000000
Lock:     .word 0x00000000 ; #0 means unlocked

   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
fail_enter:
   ldr    r0, [r3]
   cbnz   r0, fail_enter  ; if locked
   ldr    r2, [r4]
   add    r2, #1
   str    r2, [r4]
   add    r1, #1
   b      for_enter
end_for_enter:
```

Critical section

```
   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
fail_leave:
   ldr    r0, [r3]
   cbnz   r0, fail_leave  ; if locked
   ldr    r2, [r4]
   sub    r2, #1
   str    r2, [r4]
   add    r1, #1
   b      for_leave
end_for_leave:
```

Critical section

```
Count:    .word 0x00000000
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   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
fail_enter:
   ldr    r0, [r3]
   cbnz   r0, fail_enter   ; if locked
   mov    r0, #1           ; lock value
   str    r0, [r3]         ; lock

   ldr    r2, [r4]
   add    r2, #1
   str    r2, [r4]

   add    r1, #1
   b      for_enter
end_for_enter:
```

Critical section

```
   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
fail_leave:
   ldr    r0, [r3]
   cbnz   r0, fail_leave   ; if locked
   mov    r0, #1           ; lock value
   str    r0, [r3]         ; lock

   ldr    r2, [r4]
   sub    r2, #1
   str    r2, [r4]

   add    r1, #1
   b      for_leave
end_for_leave:
```

Critical section

```
Count:    .word 0x00000000
Lock:     .word 0x00000000 ; #0 means unlocked

   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
fail_enter:
   ldrex  r0, [r3]
   cbnz   r0, fail_enter   ; if locked
   mov    r0, #1           ; lock value
   strex  r0, [r3]         ; try lock
   cbnz   r0, fail_enter   ; if touched
   dmb                     ; sync memory
   ldr    r2, [r4]
   add    r2, #1
   str    r2, [r4]
   add    r1, #1
   b      for_enter
end_for_enter:
```

```
   ldr    r3, =Lock
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   sub    r2, #1
   str    r2, [r4]
   add    r1, #1
   b      for_leave
end_for_leave:
```

Any context switch needs to clear reservations

Critical section

Critical section

```
Count:    .word 0x00000000
Lock:     .word 0x00000000 ; #0 means unlocked

   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_enter:
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   add    r2, #1
   str    r2, [r4]
   dmb                     ; sync memory
   mov    r0, #0           ; unlock value
   str    r0, [r3]         ; unlock
   add    r1, #1
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```

Any context switch needs to clear reservations

Critical section

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   ldr    r3, =Lock
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   dmb                     ; sync memory
   mov    r0, #0           ; unlock value
   str    r0, [r3]         ; unlock
   add    r1, #1
   b      for_leave
end_for_leave:
```

Critical section

Mutual exclusion: atomic test-and-set operation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

```
task body Pi is
L : Flag;
begin
 loop
  loop
   [L := C; C := 1];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_i;
  C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag;
begin
 loop
  loop
   [L := C; C := 1];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  C := 0;
 end loop;
end Pj;
```

☞ Does that work?

Mutual exclusion: atomic test-and-set operation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

```
task body Pi is
L : Flag;
begin
 loop
  loop
   [L := C; C := 1];
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   ------ change process
  end loop;
  ------ critical_section_i;
  C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag;
begin
 loop
  loop
   [L := C; C := 1];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  C := 0;
 end loop;
end Pj;
```

☞ Mutual exclusion!, No deadlock!, No global live-lock!

☞ Works for any dynamic number of processes.

☞ Individual starvation possible! Busy waiting loops!

Mutual exclusion: atomic exchange operation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

```
task body Pi is
L : Flag := 1;
begin
 loop
  loop
   [Temp := L; L := C; C := Temp];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_i;
  L := 1; C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag := 1;
begin
 loop
  loop
   [Temp := L; L := C; C := Temp];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  L := 1; C := 0;
 end loop;
end Pj;
```

☞ Does that work?

Mutual exclusion: atomic exchange operation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

```
task body Pi is
L : Flag := 1;
begin
 loop
  loop
   [Temp := L; L := C; C := Temp];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_i;
  L := 1; C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag := 1;
begin
 loop
  loop
   [Temp := L; L := C; C := Temp];
   exit when L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  L := 1; C := 0;
 end loop;
end Pj;
```

- ☞ Mutual exclusion!, No deadlock!, No global live-lock!
- ☞ Works for any dynamic number of processes.
- ☞ Individual starvation possible! Busy waiting loops!

Mutual exclusion: memory cell reservation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

```
task body Pi is
L : Flag;
begin
 loop
  loop
   L :R= C; C :T= 1;
   exit when Untouched and L = 0;
   ------ change process
  end loop;
  ------ critical_section_i;
  C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag;
begin
 loop
  loop
   L :R= C; C :T= 1;
   exit when Untouched and L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  C := 0;
 end loop;
end Pj;
```

☞ Does that work?

Mutual exclusion: memory cell reservation

```
type Flag is Natural range 0..1; C : Flag := 0;
```

Any context switch needs to clear reservations

```
task body Pi is
L : Flag;
begin
 loop
  loop
   L :R= C; C :T= 1;
   exit when Untouched and L = 0;
   ------ change process
  end loop;
  ------ critical_section_i;
  C := 0;
 end loop;
end Pi;
```

```
task body Pj is
L : Flag;
begin
 loop
  loop
   L :R= C; C :T= 1;
   exit when Untouched and L = 0;
   ------ change process
  end loop;
  ------ critical_section_j;
  C := 0;
 end loop;
end Pj;
```

☞ Mutual exclusion!, No deadlock!, No global live-lock!

☞ Works for any dynamic number of processes.

☞ Individual starvation possible! Busy waiting loops!

```
Count:    .word 0x00000000
Lock:     .word 0x00000000 ; #0 means unlocked

   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
fail_enter:
   ldrex  r0, [r3]
   cbnz   r0, fail_enter   ; if locked
   mov    r0, #1           ; lock value
   strex  r0, [r3]         ; try lock
   cbnz   r0, fail_enter   ; if touched
   dmb                     ; sync memory
   ldr    r2, [r4]
   add    r2, #1
   str    r2, [r4]
   dmb                     ; sync memory
   mov    r0, #0           ; unlock value
   str    r0, [r3]         ; unlock
   add    r1, #1
   b      for_enter
end_for_enter:
```

Any context switch needs to clear reservations

Critical section

```
   ldr    r3, =Lock
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
fail_leave:
   ldrex  r0, [r3]
   cbnz   r0, fail_leave   ; if locked
   mov    r0, #1           ; lock value
   strex  r0, [r3]         ; try lock
   cbnz   r0, fail_leave   ; if touched
   dmb                     ; sync memory
   ldr    r2, [r4]
   sub    r2, #1
   str    r2, [r4]
   dmb                     ; sync memory
   mov    r0, #0           ; unlock value
   str    r0, [r3]         ; unlock
   add    r1, #1
   b      for_leave
end_for_leave:
```

Asks for permission

Critical section

Mutual exclusion

```
Count: .word 0x00000000

    ldr   r4, =Count
    mov   r1, #1
for_enter:
    cmp   r1, #100
    bgt   end_for_enter
enter_strex_fail:
    ldrex r2, [r4] ; tag [r4] as exclusive
    add   r2, #1
    strex r2, [r4] ; only if untouched
    cbnz  r2, enter_strex_fail
    add   r1, #1
    b     for_enter
end_for_enter:
```

```
    ldr   r4, =Count
    mov   r1, #1
for_leave:
    cmp   r1, #100
    bgt   end_for_leave
leave_strex_fail:
    ldrex r2, [r4] ; tag [r4] as exclusive
    sub   r2, #1
    strex r2, [r4] ; only if untouched
    cbnz  r2, leave_strex_fail
    add   r1, #1
    b     for_leave
end_for_leave:
```

Any context switch needs to clear reservations

Asks for forgiveness

☞ Light weight solution – sometimes referred to as “lock-free” or “lockless”.

Beyond atomic hardware operations

Semaphores

Basic definition (Dijkstra 1968)

Assuming the following three conditions on a shared memory cell between processes:

- a set of processes agree on a variable **S** operating as a flag to indicate synchronization conditions
- an atomic operation **P** on S — for 'passeren' (Dutch for 'pass'):

  P(S): `[as soon as S > 0 then S := S - 1]` ☞ this is a potentially delaying operation
- an atomic operation **V** on S — for 'vrygeven' (Dutch for 'to release'):

  V(S): `[S := S + 1]`

☞ *then* the variable **S** is called a **Semaphore**.

Beyond atomic hardware operations

Semaphores

… as supplied by operating systems and runtime environments

- a set of processes $P_1 \ldots P_N$ agree on a variable **S** operating as a flag to indicate synchronization conditions

- an atomic operation **Wait** on S: (aka 'Suspend_Until_True', 'sem_wait', …)

  Process P_i : **Wait** (S):

  ```
  [if S > 0 then S := S - 1
            else suspend Pi on S]
  ```

- an atomic operation **Signal** on S: (aka 'Set_True', 'sem_post', …)

  Process P_i : **Signal** (S):

  ```
  [if ∃ Pj suspended on S then release Pj
                          else S := S + 1]
  ```

☞ *then* the variable **S** is called a **Semaphore** in a scheduling environment.

Semaphores

Types of semaphores:

- **Binary semaphores**: restricted to [0, 1] or [False, True] resp.
  Multiple `V` (`Signal`) calls have the same effect than a single call.
  - Atomic hardware operations support binary semaphores.
  - Binary semaphores are sufficient to create all other semaphore forms.
- **General semaphores** (counting semaphores): non-negative number; (range limited by the system) `P` and `V` increment and decrement the semaphore by one.
- **Quantity semaphores**: The increment (and decrement) value for the semaphore is specified as a parameter with P and V.

☞ All types of semaphores must be initialized:
often the number of processes which are allowed inside a critical section, i.e. '1'.

```
Count:    .word 0x00000000
Sema:     .word 0x00000001

   ldr    r3, =Sema
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
wait_1:
   ldr    r0, [r3]
   cbz    r0, wait_1 ; if Semaphore = 0
   …                                    Critical section
   add    r1, #1
   b      for_enter
end_for_enter:
```

```
   ldr    r3, =Sema
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
wait_2:
   ldr    r0, [r3]
   cbz    r0, wait_2 ; if Semaphore = 0
   …                                    Critical section
   add    r1, #1
   b      for_leave
end_for_leave:
```

```
Count:    .word 0x00000000
Sema:     .word 0x00000001

    ldr   r3, =Sema
    ldr   r4, =Count
    mov   r1, #1
for_enter:
    cmp   r1, #100
    bgt   end_for_enter
wait_1:
    ldr   r0, [r3]
    cbz   r0, wait_1 ; if Semaphore = 0
    sub   r0, #1     ; dec Semaphore
    str   r0, [r3]   ; update

    …                                   Critical section

    add   r1, #1
    b     for_enter
end_for_enter:
```

```
    ldr   r3, =Sema
    ldr   r4, =Count
    mov   r1, #1
for_leave:
    cmp   r1, #100
    bgt   end_for_leave
wait_2:
    ldr   r0, [r3]
    cbz   r0, wait_2 ; if Semaphore = 0
    sub   r0, #1     ; dec Semaphore
    str   r0, [r3]   ; update

    …                                   Critical section

    add   r1, #1
    b     for_leave
end_for_leave:
```

```
Count:    .word 0x00000000
Sema:     .word 0x00000001

   ldr    r3, =Sema
   ldr    r4, =Count
   mov    r1, #1
for_enter:
   cmp    r1, #100
   bgt    end_for_enter
wait_1:
   ldrex  r0, [r3]
   cbz    r0, wait_1 ; if Semaphore = 0
   sub    r0, #1     ; dec Semaphore
   strex  r0, [r3]   ; try update
   cbnz   r0, wait_1 ; if touched
   dmb               ; sync memory
   …
   add    r1, #1
   b      for_enter
end_for_enter:
```

```
   ldr    r3, =Sema
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
wait_2:
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   cbz    r0, wait_2 ; if Semaphore = 0
   sub    r0, #1     ; dec Semaphore
   strex  r0, [r3]   ; try update
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   …
   add    r1, #1
   b      for_leave
end_for_leave:
```

Any context switch needs to clear reservations

Critical section

Critical section

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Count:    .word 0x00000000
Sema:     .word 0x00000001

   ldr    r3, =Sema
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   mov    r1, #1
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   cbnz   r0, wait_1 ; if touched
   dmb               ; sync memory
   …

   ldr    r0, [r3]
   add    r0, #1         ; inc Semaphore
   str    r0, [r3]       ; update


   add    r1, #1
   b      for_enter
end_for_enter:
```

```
   ldr    r3, =Sema
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   mov    r1, #1
for_leave:
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   dmb               ; sync memory
   …

   ldr    r0, [r3]
   add    r0, #1         ; inc Semaphore
   str    r0, [r3]       ; update


   add    r1, #1
   b      for_leave
end_for_leave:
```

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   mov    r1, #1
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   ldrex  r0, [r3]
   cbz    r0, wait_1 ; if Semaphore = 0
   sub    r0, #1     ; dec Semaphore
   strex  r0, [r3]   ; try update
   cbnz   r0, wait_1 ; if touched
   dmb               ; sync memory
   …
signal_1:
   ldrex  r0, [r3]
   add    r0, #1         ; inc Semaphore
   strex  r0, [r3]       ; try update
   cbnz   r0, signal_1   ; if touched
   dmb                   ; sync memory
   add    r1, #1
   b      for_enter
end_for_enter:
```

Any context switch needs to clear reservations

Critical section

```
   ldr    r3, =Sema
   ldr    r4, =Count
   mov    r1, #1
for_leave:
   cmp    r1, #100
   bgt    end_for_leave
wait_2:
   ldrex  r0, [r3]
   cbz    r0, wait_2 ; if Semaphore = 0
   sub    r0, #1     ; dec Semaphore
   strex  r0, [r3]   ; try update
   cbnz   r0, wait_2 ; if touched
   dmb               ; sync memory
   …
signal_2:
   ldrex  r0, [r3]
   add    r0, #1         ; inc Semaphore
   strex  r0, [r3]       ; try update
   cbnz   r0, signal_2   ; if touched
   dmb                   ; sync memory
   add    r1, #1
   b      for_leave
end_for_leave:
```

Critical section

Semaphores

```
S : Semaphore := 1;
```

```
task body Pi is
begin
 loop
  ------ non_critical_section_i;
  wait (S);
   ------ critical_section_i;
  signal (S);
 end loop;
end Pi;
```

```
task body Pj is
begin
 loop
  ------ non_critical_section_j;
  wait (S);
   ------ critical_section_j;
  signal (S);
 end loop;
end Pi;
```

☞ Works?

Semaphores

```
S : Semaphore := 1;
```

```
task body Pi is
begin
 loop
  ------ non_critical_section_i;
  wait (S);
   ------ critical_section_i;
  signal (S);
 end loop;
end Pi;
```

```
task body Pj is
begin
 loop
  ------ non_critical_section_j;
  wait (S);
   ------ critical_section_j;
  signal (S);
 end loop;
end Pi;
```

- ☞ Mutual exclusion!, No deadlock!, No global live-lock!
- ☞ Works for any dynamic number of processes
- ☞ Individual starvation possible!

Semaphores

```
S1, S2 : Semaphore := 1;
```

```
task body Pi is
begin
 loop
  ------ non_critical_section_i;
  wait (S1);
  wait (S2);
   ------ critical_section_i;
  signal (S2);
  signal (S1);
 end loop;
end Pi;
```

```
task body Pj is
begin
 loop
  ------ non_critical_section_j;
  wait (S2);
  wait (S1);
   ------ critical_section_j;
  signal (S1);
  signal (S2);
 end loop;
end Pi;
```

☞ Works too?

Semaphores

```
S1, S2 : Semaphore := 1;
```

```
task body Pi is
begin
 loop
  ------ non_critical_section_i;
  wait (S1);
  wait (S2);
   ------ critical_section_i;
  signal (S2);
  signal (S1);
 end loop;
end Pi;
```

```
task body Pj is
begin
 loop
  ------ non_critical_section_j;
  wait (S2);
  wait (S1);
   ------ critical_section_j;
  signal (S1);
  signal (S2);
 end loop;
end Pi;
```

- ☞ Mutual exclusion!, No global live-lock!
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Semaphores

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S1, S2 : Semaphore := 1;
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task body Pi is
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  wait (S1);
  wait (S2);
   ------ critical_section_i;
  signal (S2);
  signal (S1);
 end loop;
end Pi;
```

```
task body Pj is
begin
 loop
  ------ non_critical_section_j;
  wait (S2);
  wait (S1);
   ------ critical_section_j;
  signal (S1);
  signal (S2);
 end loop;
end Pi;
```

- ☞ Mutual exclusion!, No global live-lock!
- ☞ Works for any dynamic number of processes.
- ☞ Individual starvation possible!
- ☞ Deadlock possible!

Concurrent programming languages offer **higher abstraction** and **safer** synchronization mechanisms.

Summary

Aynchronism

- **Interrupts & Exceptions**
  - Concept
  - Hardware/Software interaction
  - Recursive interrupts
- **Concurrency & Synchronization**
  - Race conditions
  - Synchronization
  - Passing data